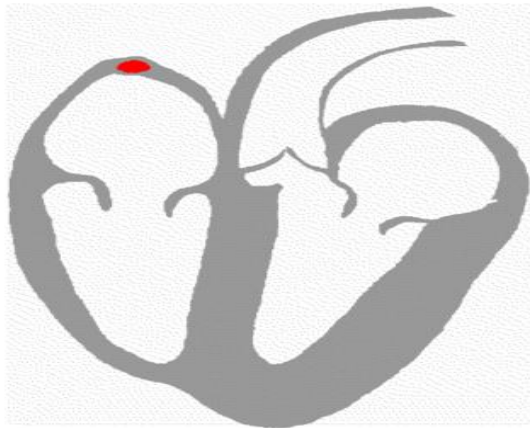


Normal Electrocardiography

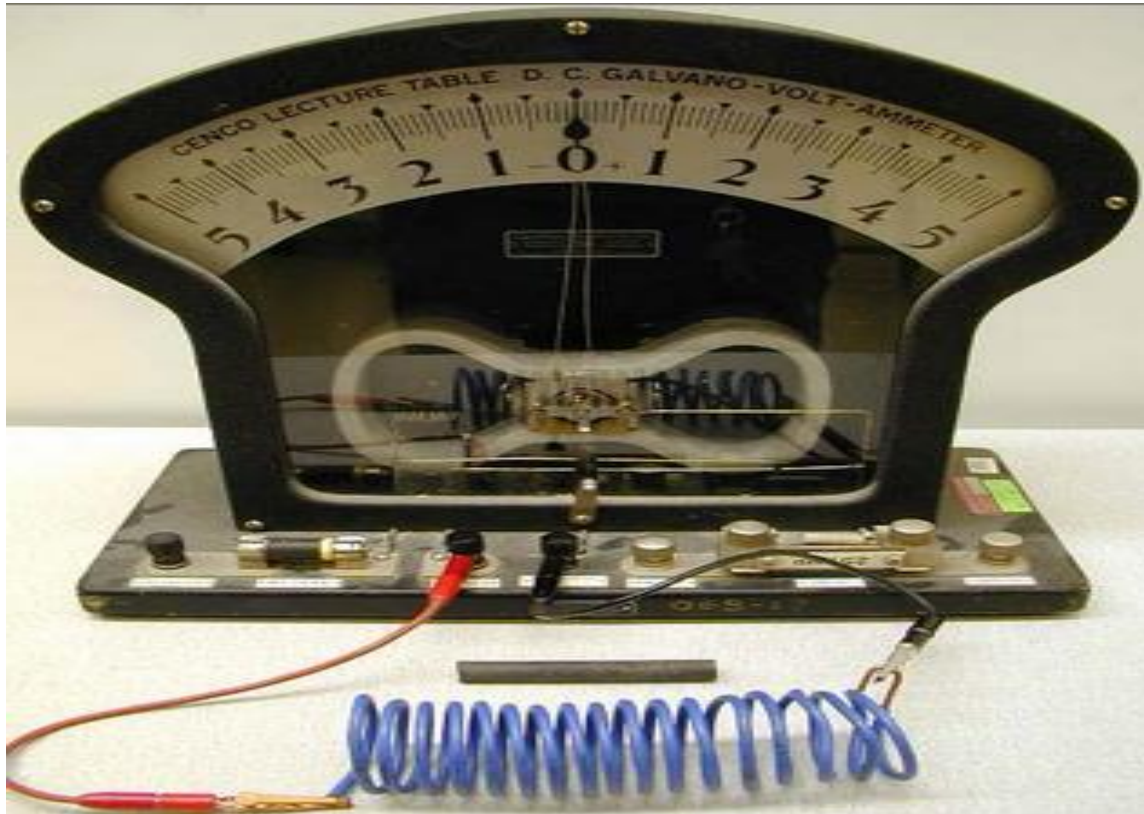


Electrocardiography

- Abbreviated by ECG/EKG.
- *Is the device used to recorded the electrical activity of the heart.*
- Graphical representation of the electrical activity of the heart.
 - It record the action potentials produced by all the heart muscle fibres during each heart beat.
 - An ECG integrates electrical information from electrodes placed at different locations on the body surface.
- Useful in diagnosing abnormal heart rates, arrhythmias, and damage of the heart.
- It works under the principle of Galvanometer.

Electrocardiography

- **Galvanometer** is an electromechanical instrument which is used for the detection of electric currents through electric circuits.



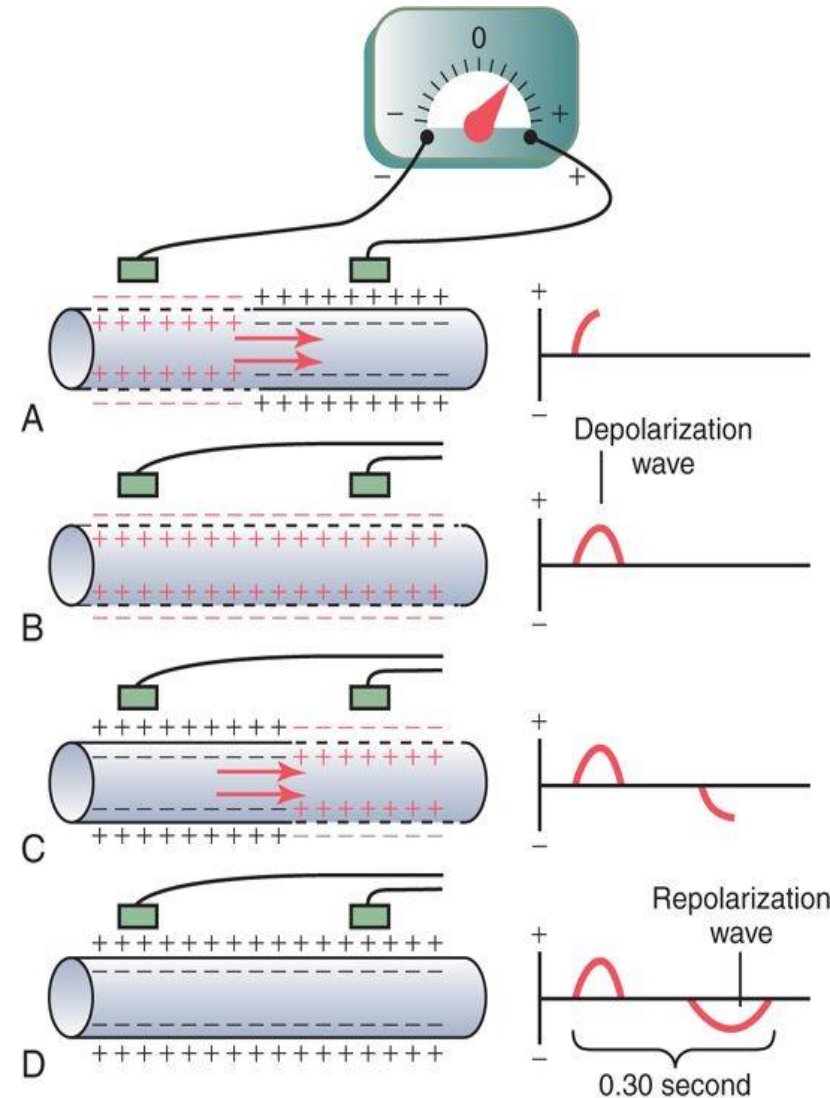
Electrocardiography

- *Basic principles of Galvanometer.*

1. As **positive current** moving toward the positive electrode, deflection is positive
2. As positive current moving away from the positive electrode, deflection is to negative.
3. When negative current moving to the negative electrodes deflection is positive.
4. Complete **depolarization** or **repolarization** of the myocardium allows the galvanometer to deflect toward **zero**.
 - B/c no electrical activity is taken place during this time.

Electrocardiography

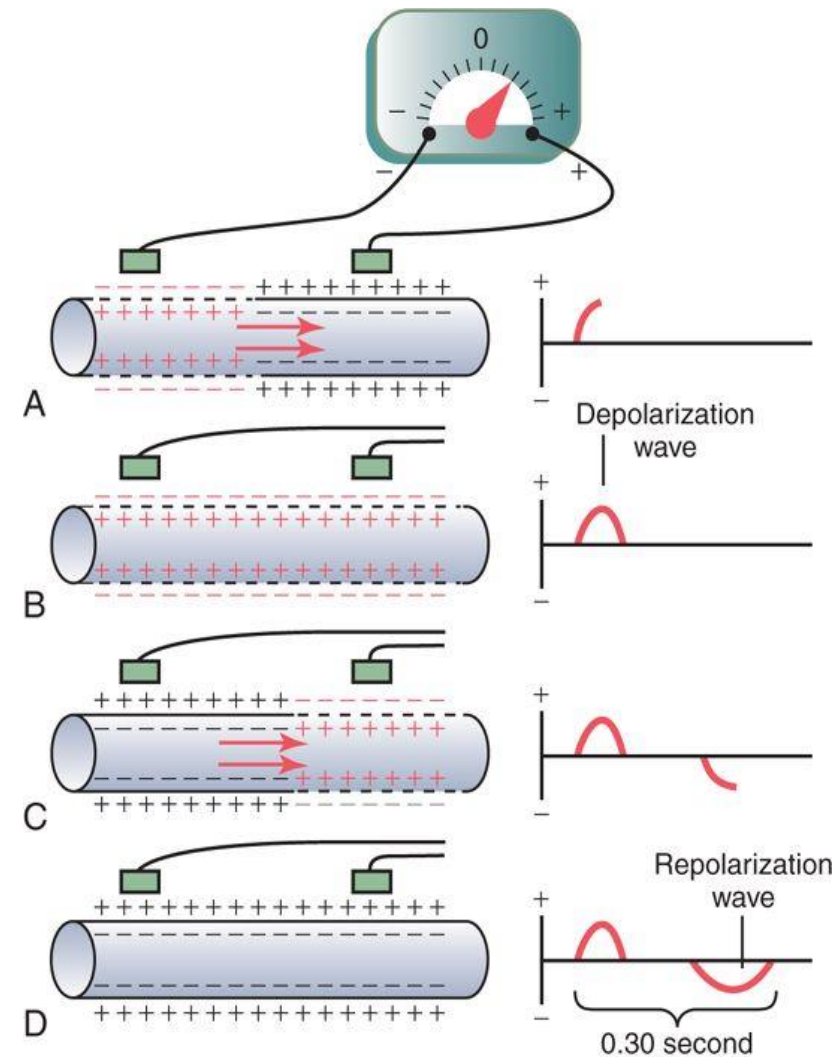
- In Figure A, **depolarization**, demonstrated by red positive charges inside and red negative charges outside, is traveling from left to right.
 - The first half of the fiber has already depolarized, while the remaining half is still polarized.
 - Therefore, the left electrode on the outside of the fiber is in an area of negativity, and the right electrode is in an area of positivity; this causes the meter to record **positively**.



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Electrocardiography

- In Figure B, depolarization has extended over the entire muscle fiber, and the recording to the right has returned to the zero baseline because both electrodes are now in areas of equal negativity.
 - *No potential is recorded in the electrocardiogram when the myocardium muscle is either completely polarized or completely depolarized*



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Flow of Electrical Currents in the Heart

- Typically, in adult the mean electrical axis lies along a line extending from the **base to the apex** of the heart.
 - During the rest of the depolarization process, current continues to flow from the endocardial surface outward through the ventricular muscle mass (in the direction of base to apex).

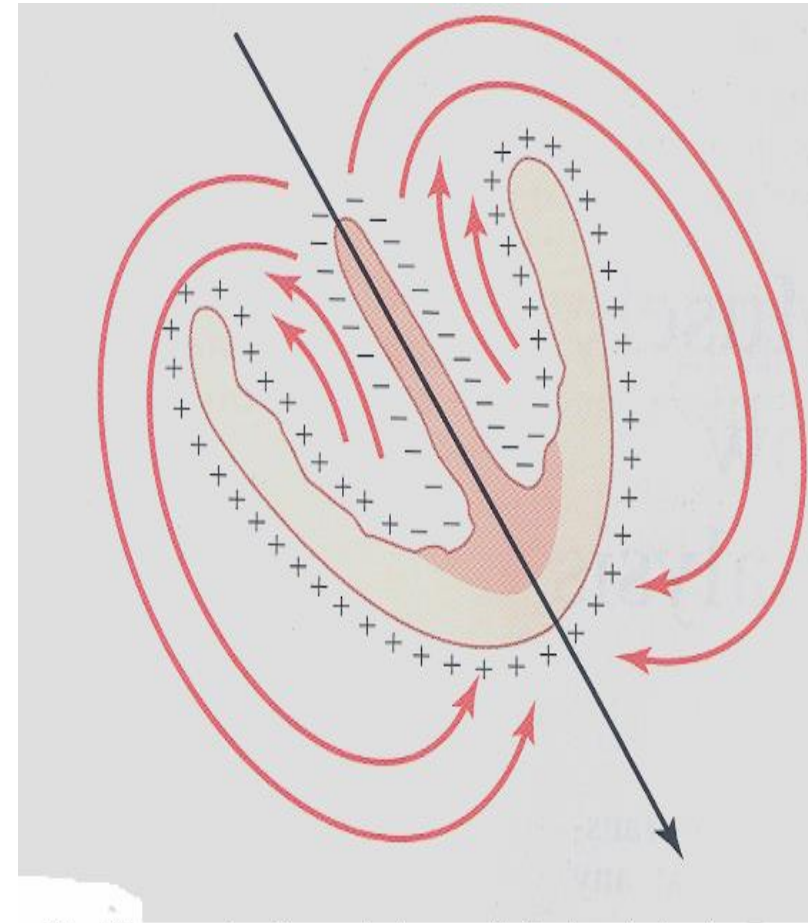


Fig..Mean vector through the partially depolarized ventricles

Flow of Electrical Currents in the Heart

- Then, immediately before depolarization has completed its course through the ventricles, the average direction of current flow (depolarization process) **reverses for about 0.01 second**.
 - Current will flow from ventricular apex toward the base (*because the last part of the heart to become depolarized is the outer walls of the ventricles near the base of the heart*).

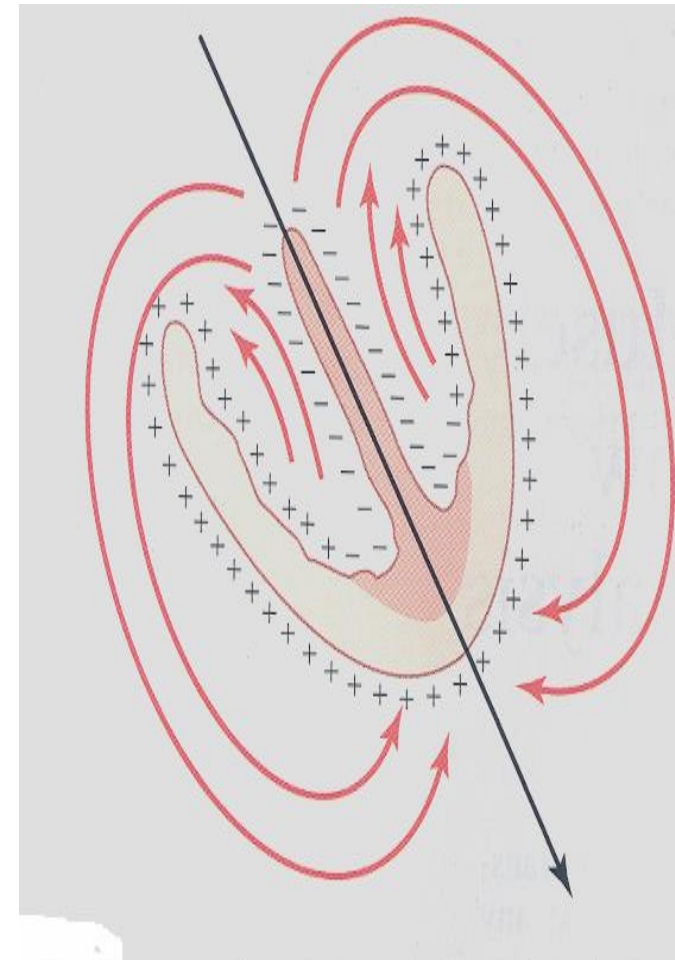
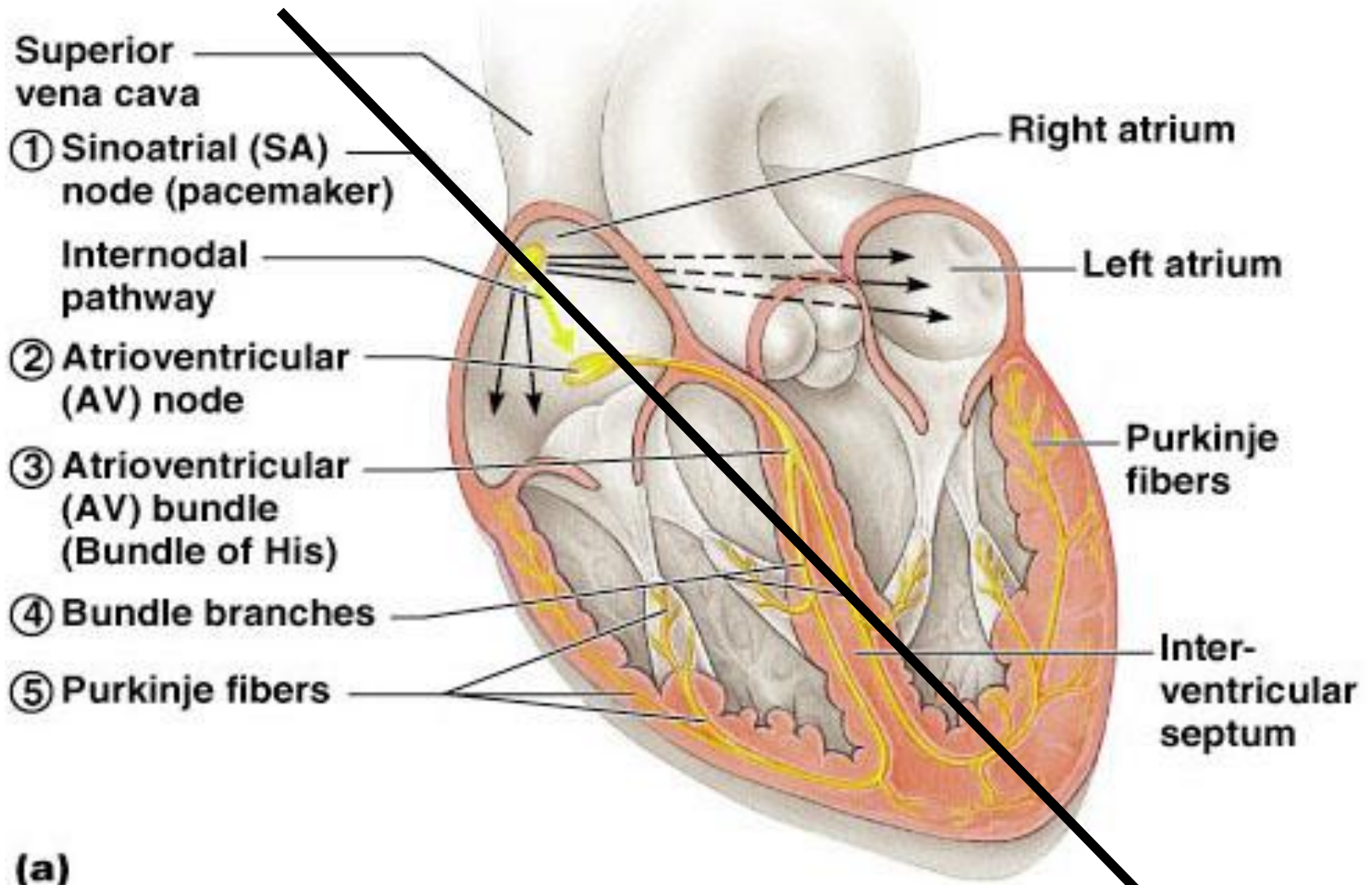


Fig. Mean vector through the partially depolarized ventricles

Heart Physiology: Sequence of Excitation

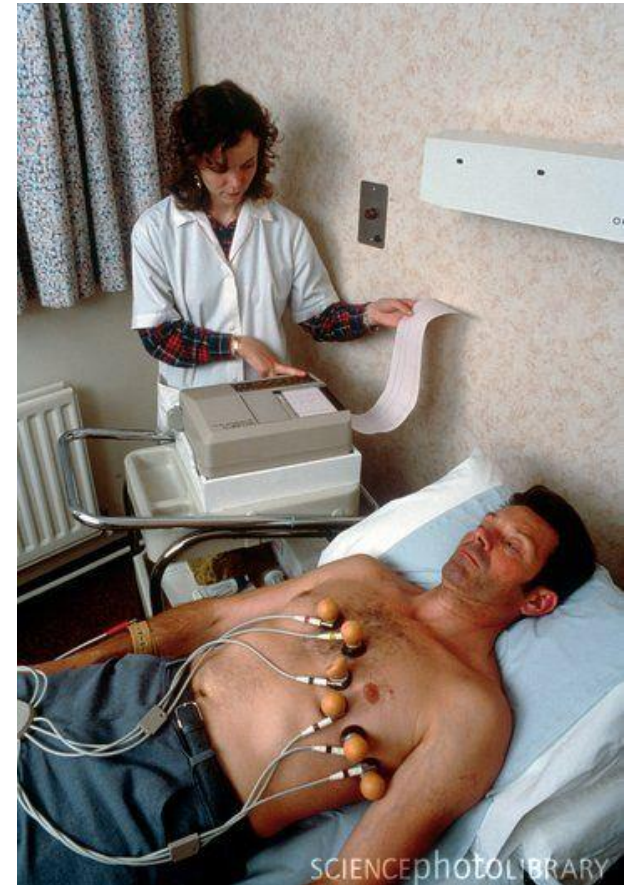


Fundamentals of Electrocardiography

- ECG is the electrical activity of the heart that is conducted through the body fluid and recorded from the surface of the body using highly sensitive electrodes.
- Recorded from specific sites of the body in graphic form **relating voltage (vertical axis) with time (horizontal axis)**.
- It is the sum of all the depolarization occurring in the cardiac muscle fibers during each cardiac cycle.
- ❖ Developed by W. Einthoven in Leiden and A. Waller in London in 1909.

ECG...cont'd

- Differences of electric potentials are recorded b/n pairs of points (**electrodes**) on the external surface of the body or b/n skin loci and indifferent or reference electrode.
- It provides events about arrhythmia, hypertrophy, myocardial damage, ischemia and necrosis
- ❖ Electrical axis of the heart is **59°**



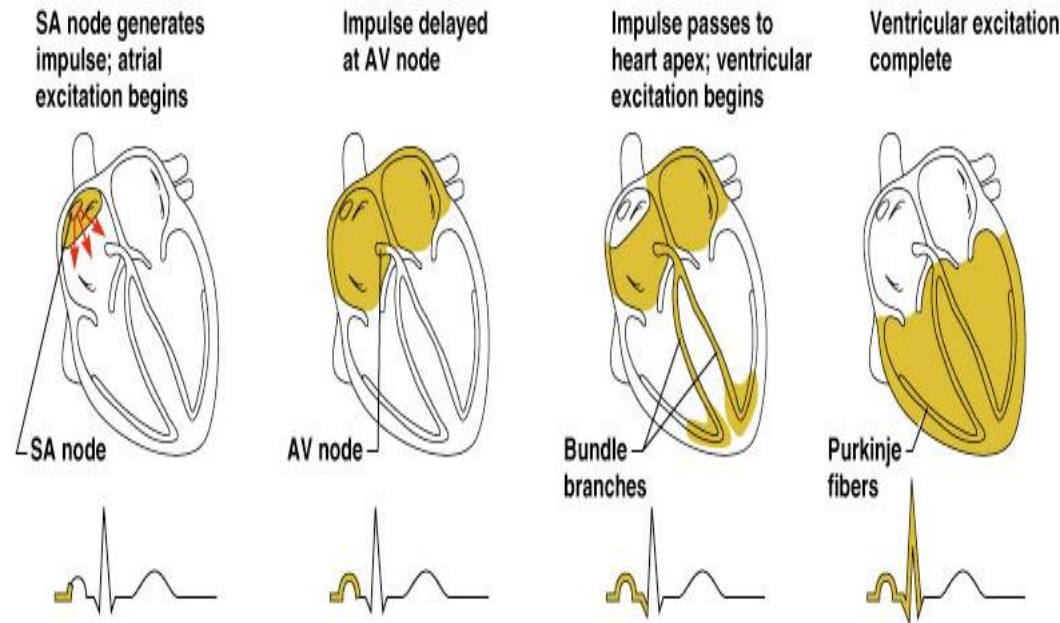
ECG...cont'd

ECG Conventions

- **Recording points** are wrist, ankle, and skin on chest.
- **Right leg** is ground (earth).
- The ECG paper speed is ordinarily **25 mm/sec**.
 - As a result, each 1 mm (small) horizontal box corresponds to 0.04 second (40 msec).
 - One larger boxes that include five small boxes and hence represent 0.20 sec (200 ms) intervals.
 - Vertically, the ECG graph measures the height (amplitude) of a given wave or deflection, as 10 mm (10 small boxes) equals 1 mV with standard calibration
 - Each small 1-mm square represents 0.1 mV in voltage.

Heart Excitation Related to ECG

❖ *ECG Fluctuations of potential that represent the algebraic sum of the action potentials of myocardial fibers recorded extra-cellularly during the cardiac cycle.*



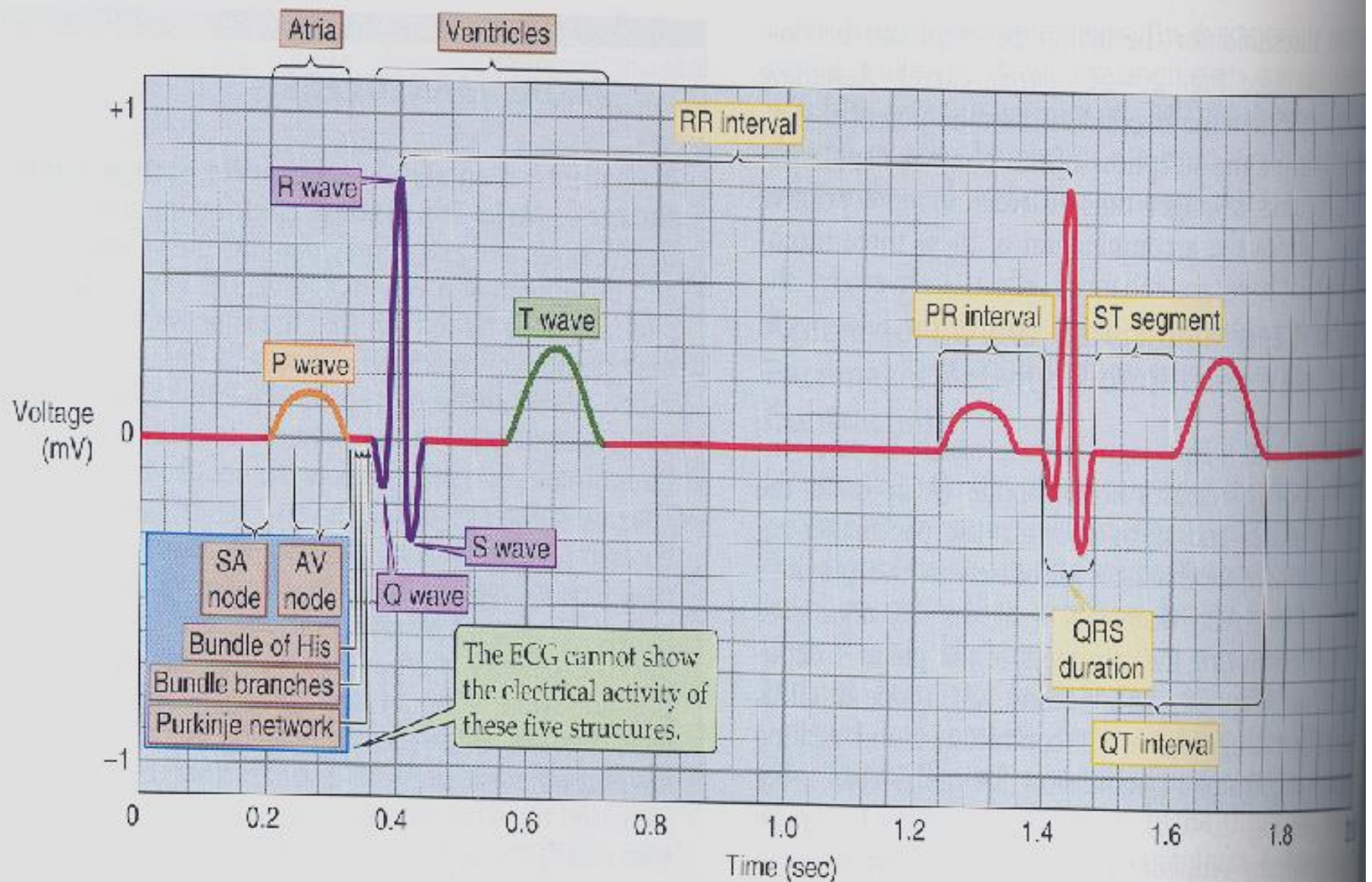
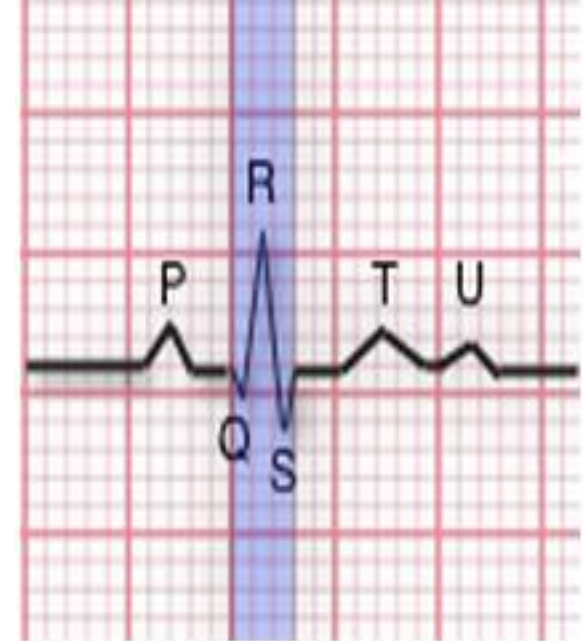


Fig. ECG



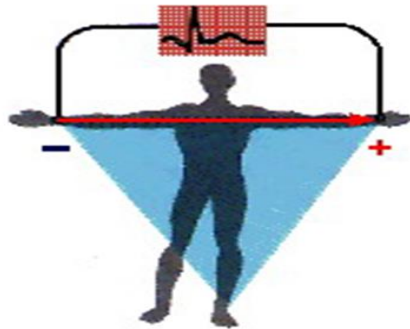
ECG Recordings

- There are 3 types of recording:
 1. Bipolar Limb Leads
 2. Augmented (Unipolar) Limb Leads
 3. Precordial (Chest) Leads

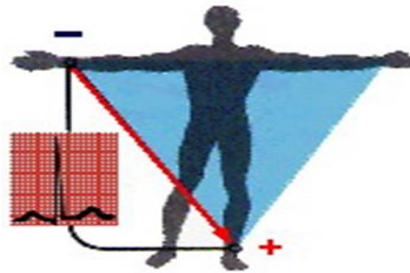
A. Bipolar Limb Leads

- Record voltage b/n two electrodes (leads) placed on the wrists and legs.
- Each lead shows waves of depolarization and repolarization: P-wave, QRS complex, T-wave and occasionally U-wave.

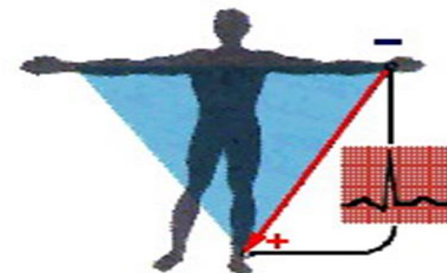
Electrocardiograms leads-Three bipolar leads



Lead I *

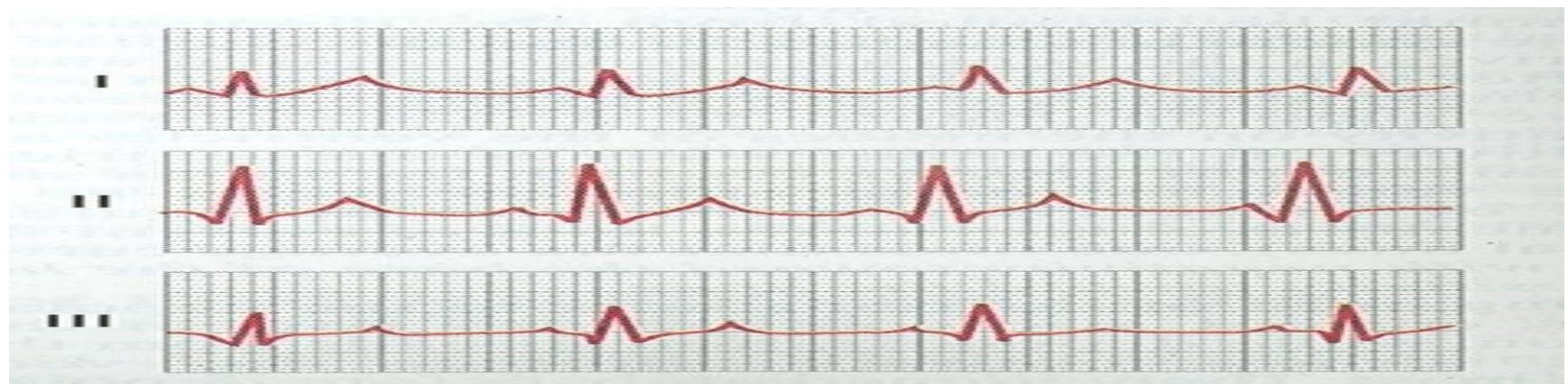


Lead II *



Lead III *

<u>Lead</u>	<u>Polarity</u>	<u>Lead Axis</u>
Lead I	right arm (-) to left arm (+)	$\pm 180^\circ - 0^\circ$
Lead II	right arm (-) to left leg (+)	$-120^\circ - +60^\circ$
Lead III	left arm (-) to left leg (+)	$-60^\circ - +120^\circ$



Bipolar Limb L...cont'd

- **Einthoven triangle:**

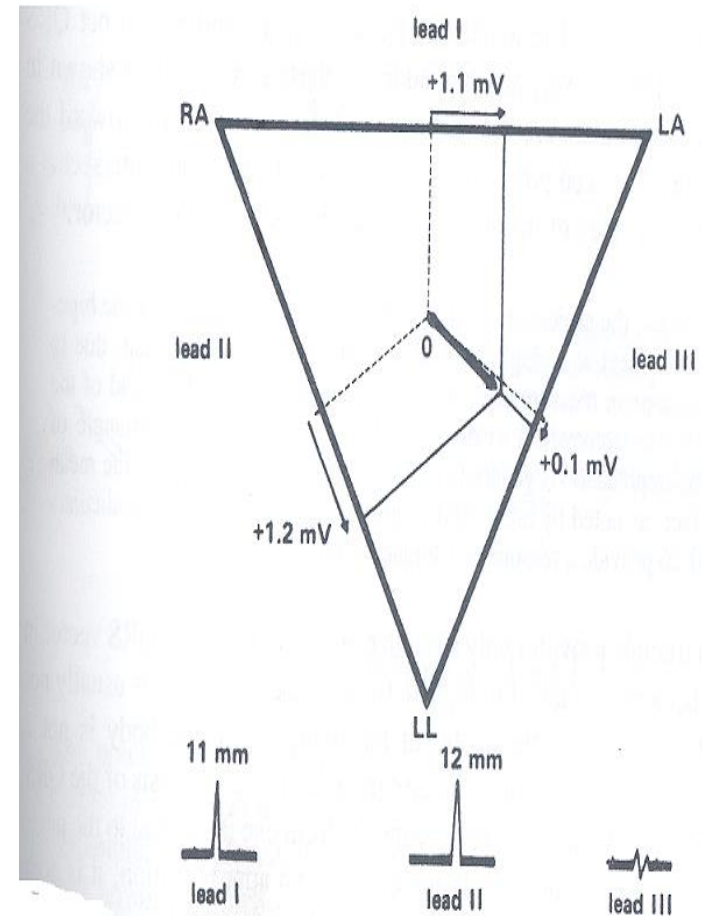
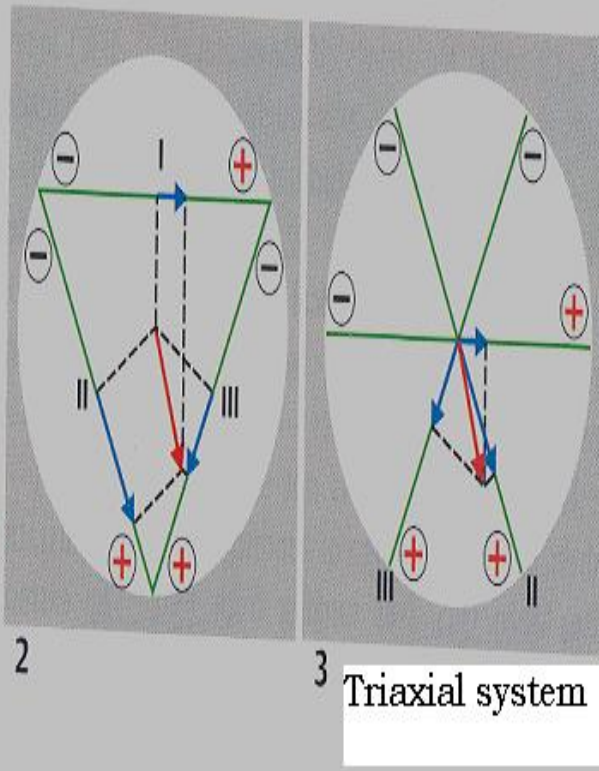
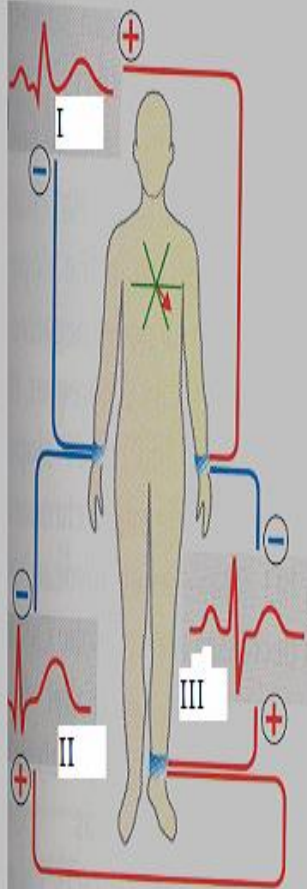
- Drawn around the heart.
- The triangle shows that the **two arms and the left leg form apices of a triangle surrounding the heart.**
- The two arms with electrodes connect electrically with the fluid around the heart and the left leg with another electrode also connects with the fluid.

- **Einthoven's law:**

- If electrical potentials of any two of the three leads are given, the 3rd one can be determined.
- The relation, $\text{Lead II} = \text{Lead I} + \text{Lead III}$, can be used to estimate electric potential of each of the leads.

Bipolar Limb L...cont'd

Einthoven leads I, II and III (bipolar)



A normal mean QRS vector projected on the Einthoven triangle, each side of which represents one of the three bipolar limb leads (I, II, III).

ECG Recordings...cont'd

B. Unipolar (Augmented) Limb Leads

- Voltage is recorded b/n a single “**exploratory electrode**” placed on the body and an electrode that is built into the electrocardiograph and maintained at zero potential (ground).
- In this system, two of the limbs are connected through electrical resistances to the **negative (indifferent) terminal** of the electrocardiograph and the 3rd limb is connected to the **positive terminal**.
- When positive terminal is placed on right arm, the lead is designated as **aVR**; when on the left arm **aVL**, and when on left leg, **aVF** (Fig. below).

Unipolar (Augmented) Limb Leads

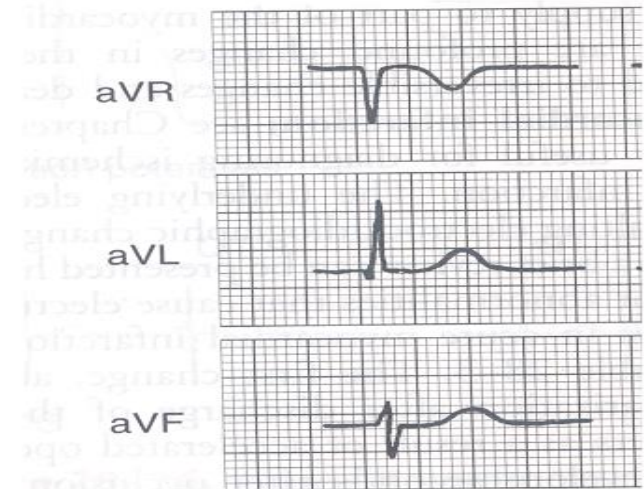
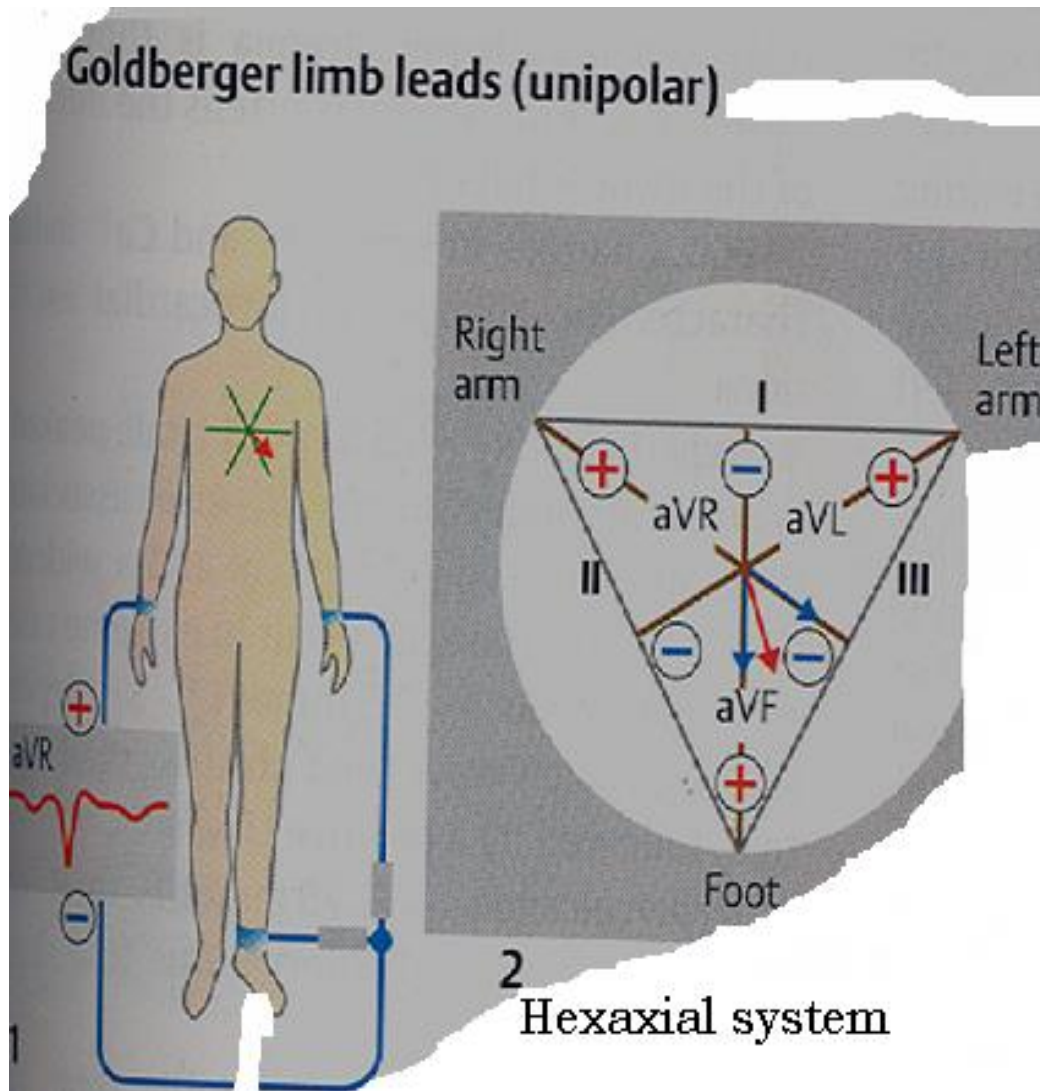


Fig..Unipolar (Goldberg) limb leads (augmented)

$$aVL: \frac{\text{Lead I} - \text{Lead III}}{2}$$

$$-aVR: \frac{\text{Lead I} + \text{Lead II}}{2}$$

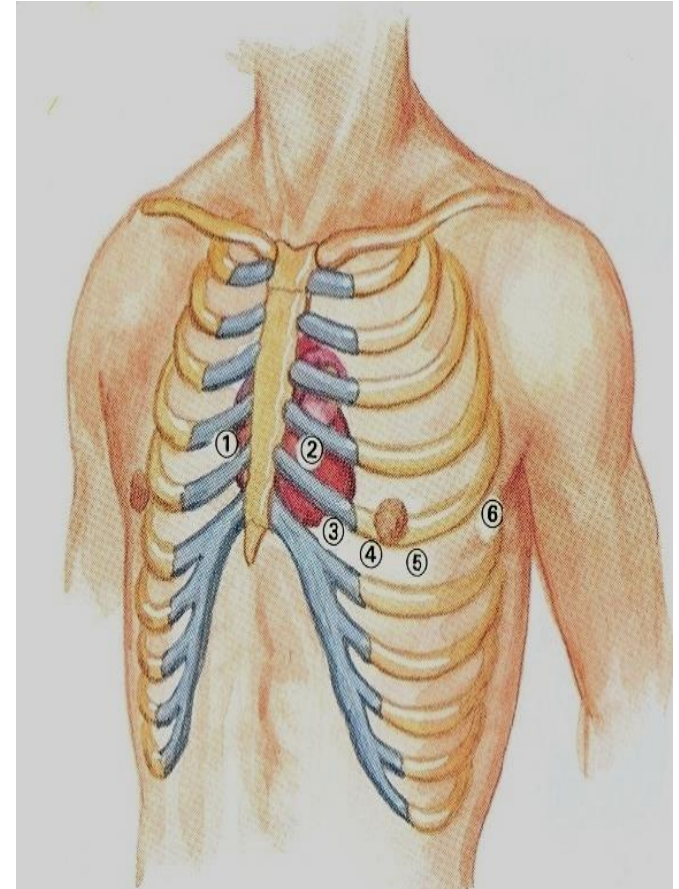
$$aVF: \frac{\text{Lead II} + \text{Lead III}}{2}$$

Goldberger's equations.

ECG Recordings...cont'd

C. Precordial (Chest) Limb Leads

- Unipolar leads labeled V1 through V6.
- One electrode is connected to the **positive terminal** of the Electrocardiograph and the **negative electrode (indifferent electrode)** is connected through equal electrical resistances to the right arm, left arm, and left leg.
- Electrodes are placed on the chest.



Precordial (Chest) Limb Leads

Standard chest leads

Lead	Position of exploring electrode
V ₁	Fourth intercostal space just to the right of the sternum
V ₂	Fourth intercostal space just to the left of the sternum
V ₃	Midway between V ₂ and V ₄
V ₄	Fifth intercostal space at the left midclavicular line
V ₅	Left anterior axillary line horizontally to the left of V ₄
V ₆	Midaxillary line horizontally to the left of V ₄ and V ₅

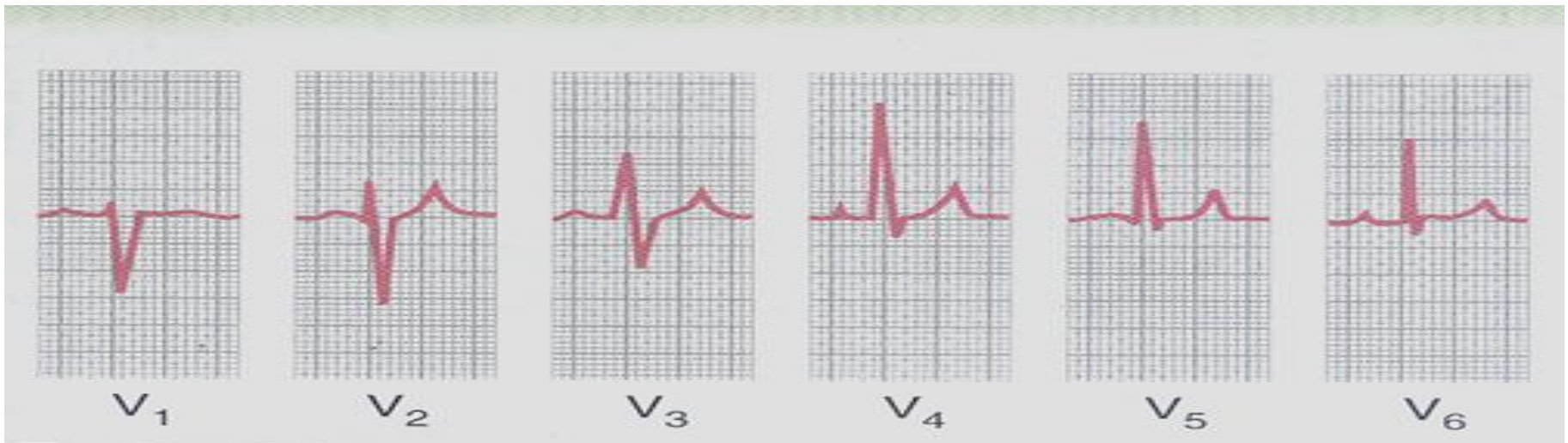


Fig. Normal ECG recorded the six chest leads

...cont'd

- Heart surfaces are close to the chest wall and each chest lead records mainly the electrical potential of the cardiac muscle immediately beneath the electrode,
 - i.e., relatively minute abnormalities in the ventricles can cause marked changes in the ECG.
- QRS in
 - V1, V2, are negative because the chest electrodes are nearer to the base of the heart (direction of electronegativity).
 - V3 is in b/n.
 - V4 - V6, are positive because they are nearer the apex (direction of electro positivity).

Axis of limb leads

- The direction from **positive** to **negative** electrode is called ***axis of the lead***.
- For example,
 - In **lead I**, the electrodes lie in the horizontal direction with the positive electrode to the left; the axis of lead I is, therefore, **0 degree**.
 - In **lead II**, electrode placed on RA connected to the torso in upper right hand corner and LL connected in the lower left-hand corner → **+60°**
 - By a similar analysis, **Lead III** → **+120°**
 - **aVR** → **210°**,
 - **aVL** → **-30°**
 - **aVF** → **+90°**

Axis of the three bipolar and three unipolar limb leads

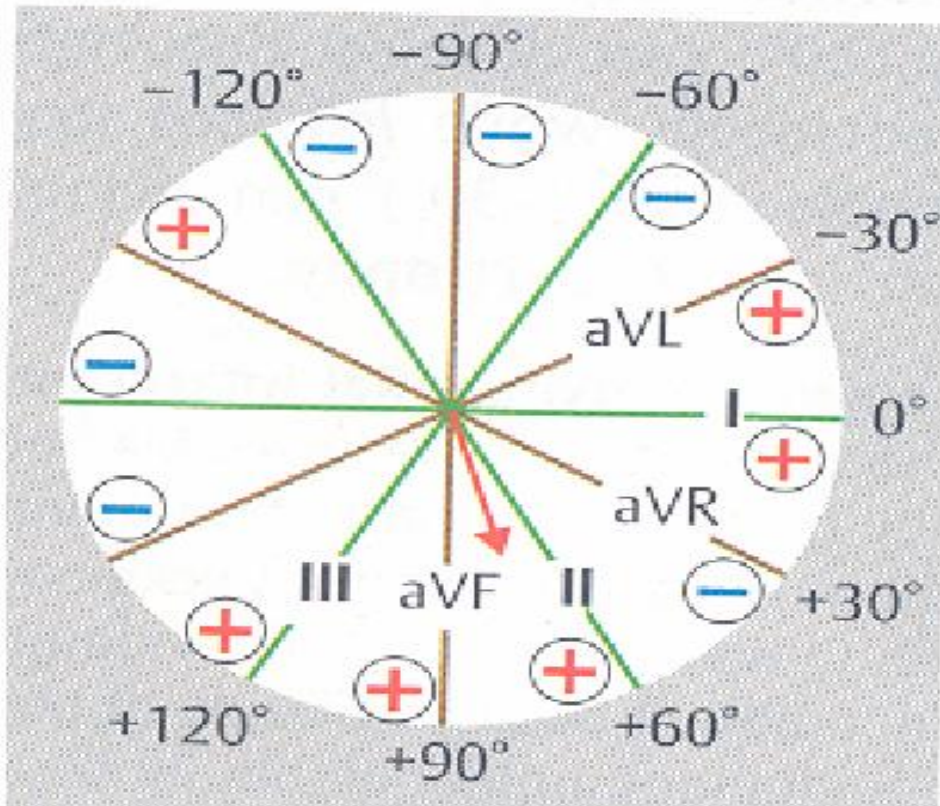


Fig..Cabrera circle

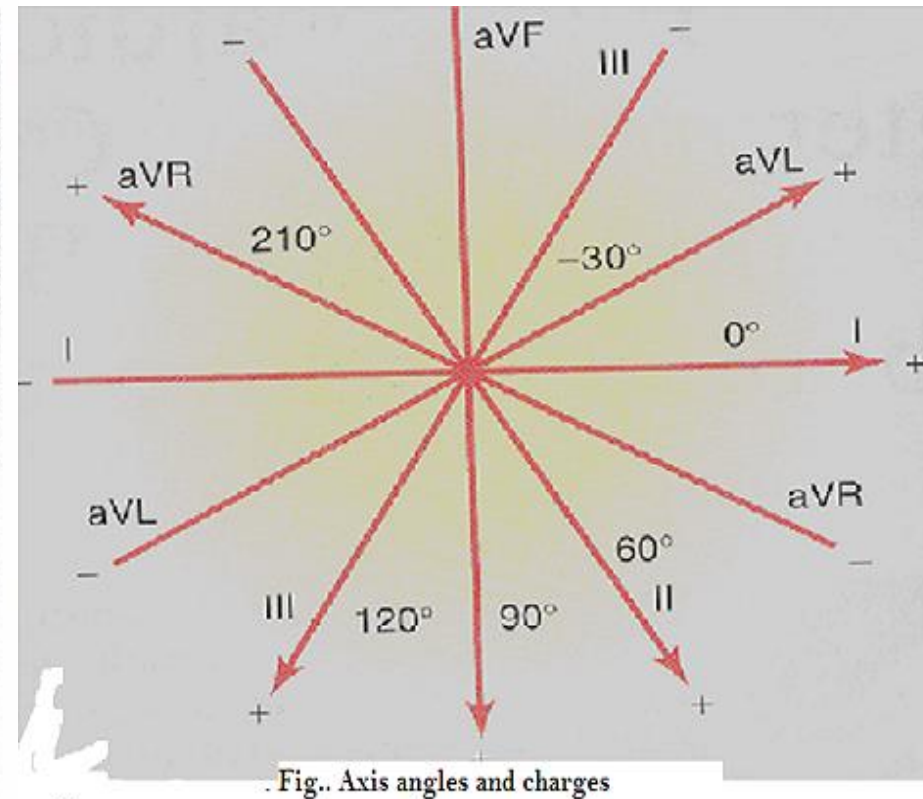


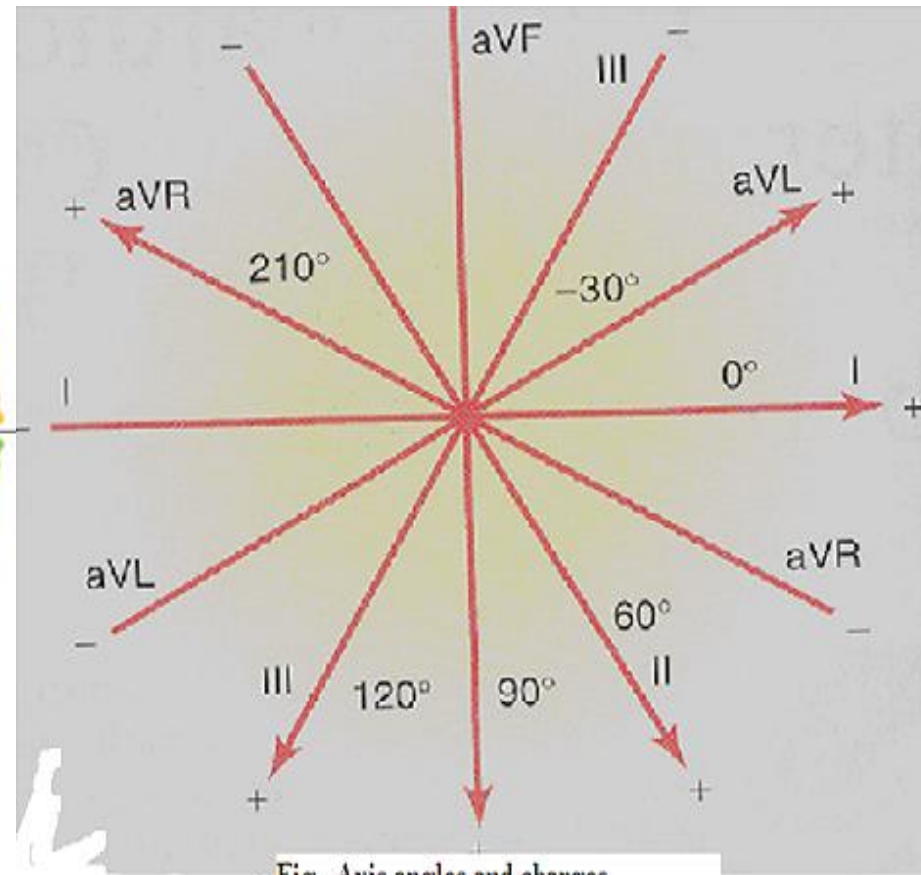
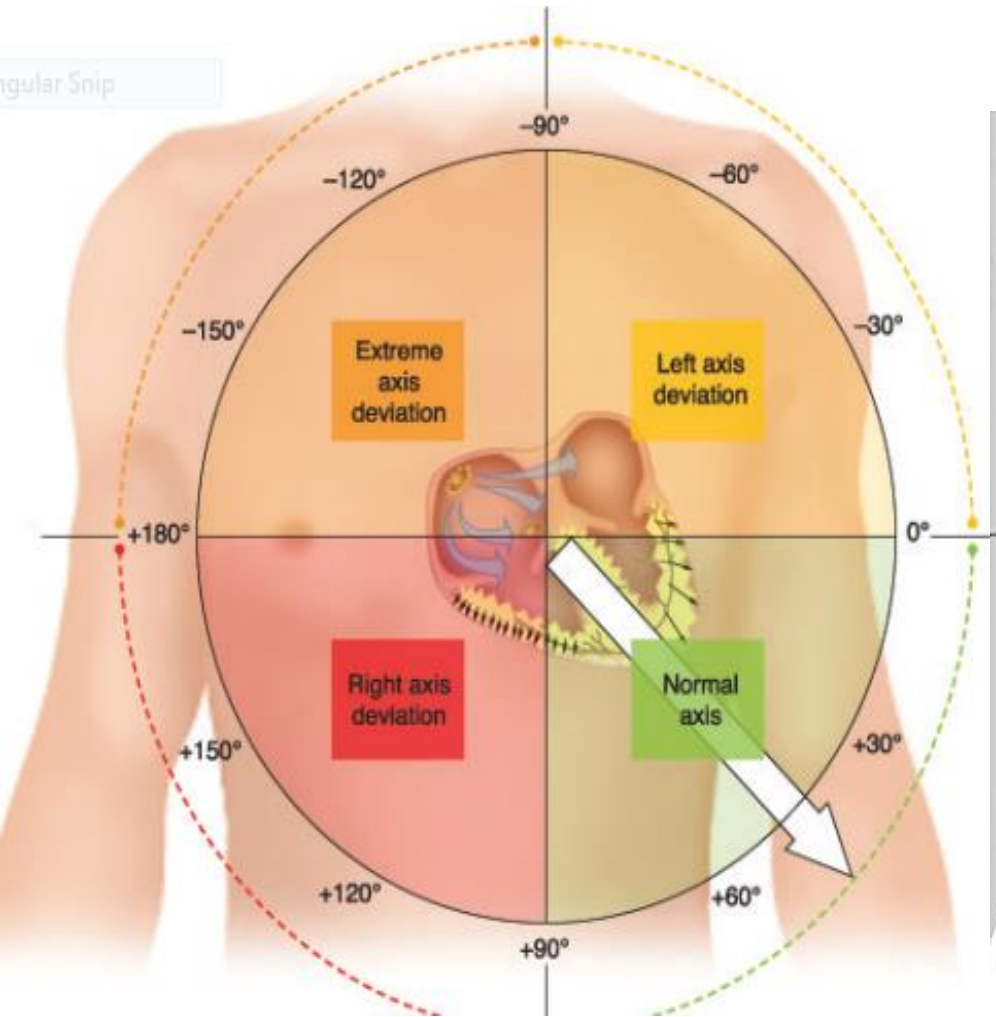
Fig.. Axis angles and charges

- The limb leads are placed on a **Cabrera Circle** with their relative axis angles and charges.

Mean Electrical Axis of the Heart

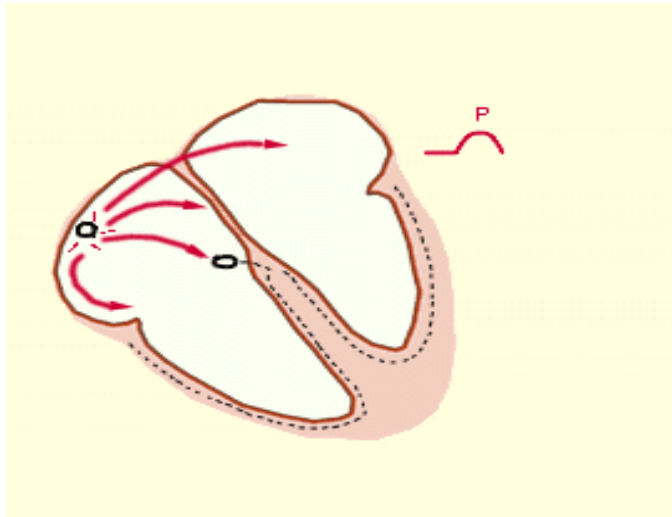
- Mean electrical axis of the ventricles = vector showing current flows from base of ventricles (-ve) toward apex (+ve).
- This amounts to 59-60°.
- Electrical axis is determined from standard limb leads
- **A positive vector** in a lead (shown in axis) causes recording in the ECG above the zero line, while **a negative vector** causes ECG recording below the zero line.

Vectorial Analysis of the Normal Electrocardiogram

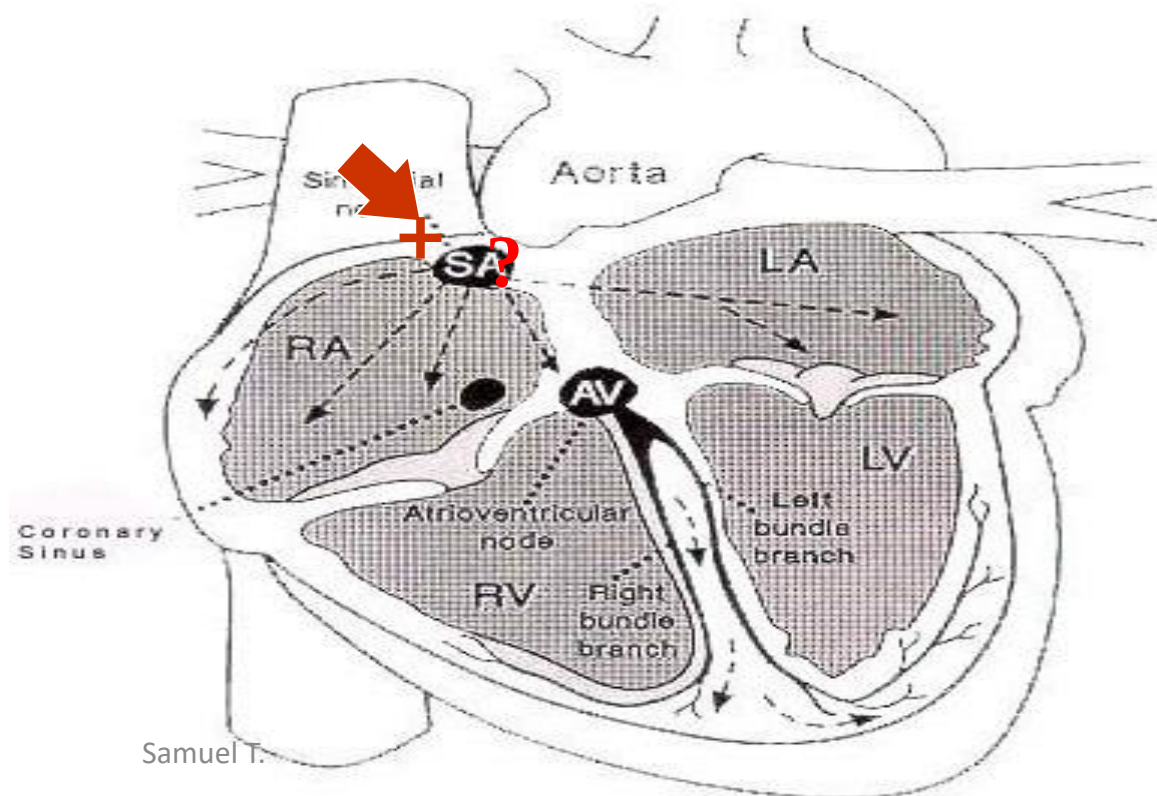


Vectorial Analysis of the Normal Electrocardiogram

- *A vector is defined as a force that can be represented by direction and magnitude.*
- Atrial depolarization from SA-node produces a down ward, left ward depolarizing vector.
- **Faster and smaller vector.**

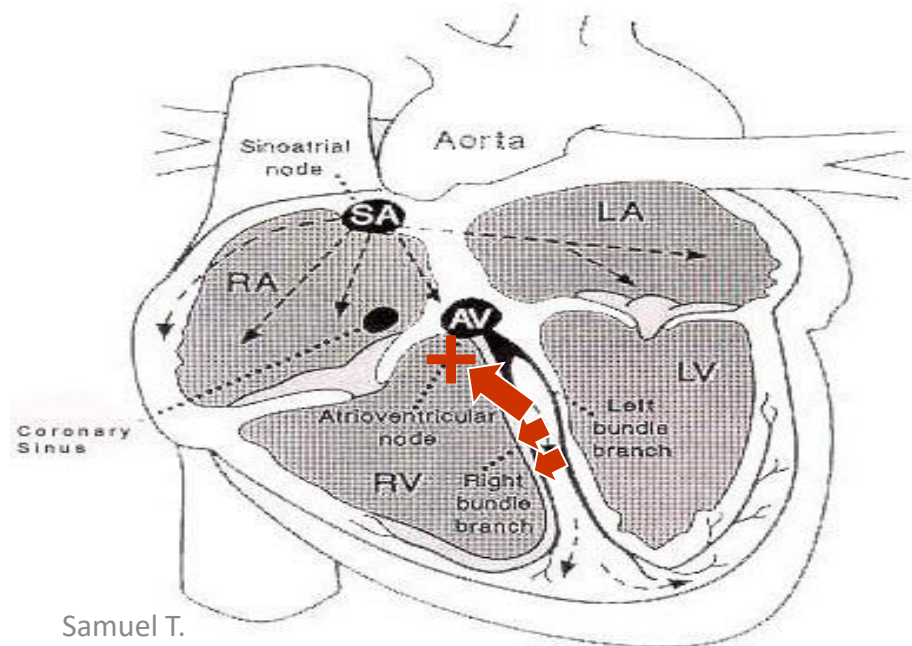
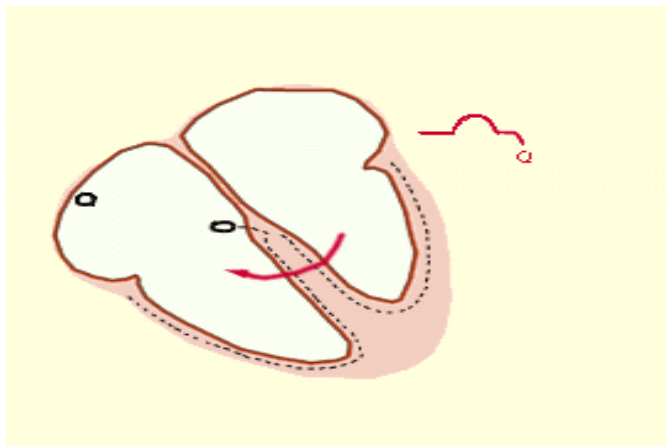


Then, AV-nodal depolarization which is not noticed in the ECG. ??????



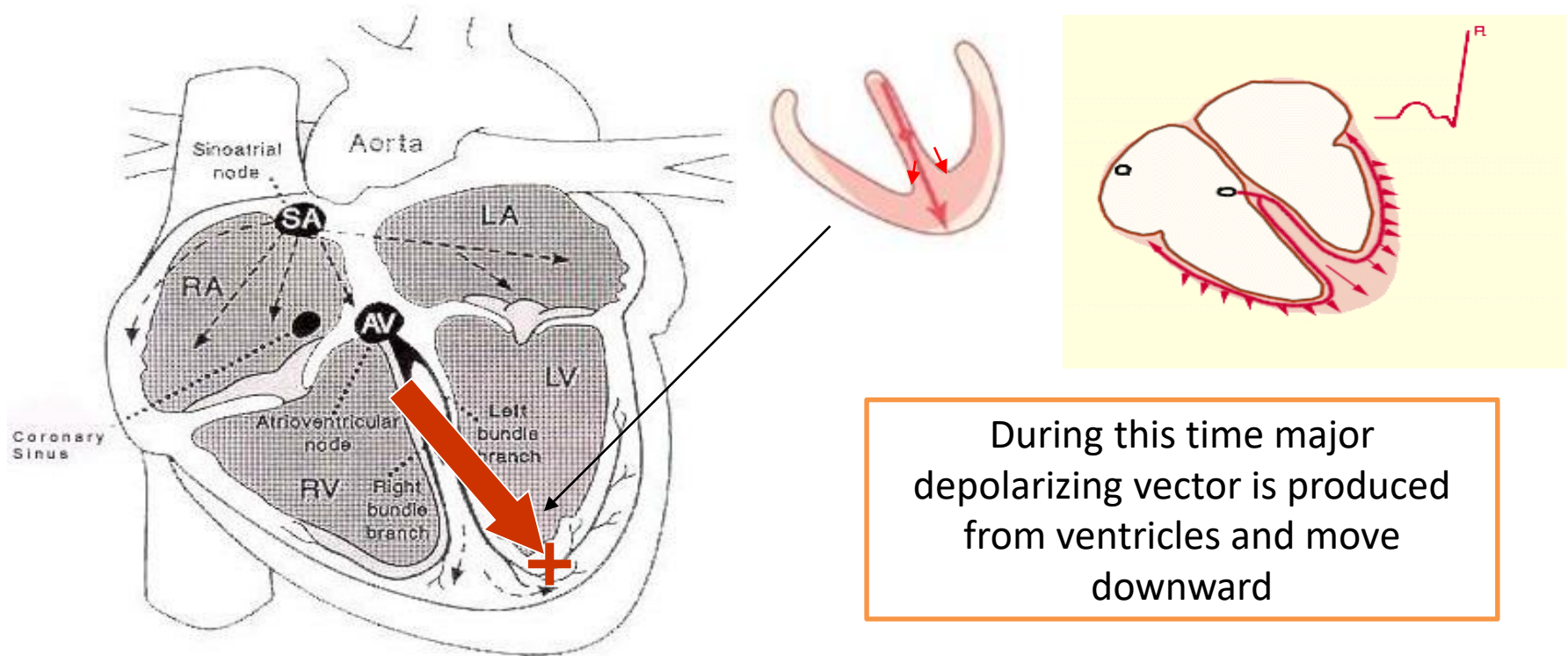
Vectorial Analysis of the Normal Electrocardiogram

- When the cardiac impulse enters the ventricles through the AV bundle, the first part of the ventricles to become depolarized *is the left endocardial surface of the septum*.
 - This depolarization vector start from the lower portion of the left septum and move **upward & right ward direction**.
 - At this time, the **vector is short/small and fast**, because only a small portion of the ventricular septum is depolarized.



Vectorial Analysis of the Normal Electrocardiogram

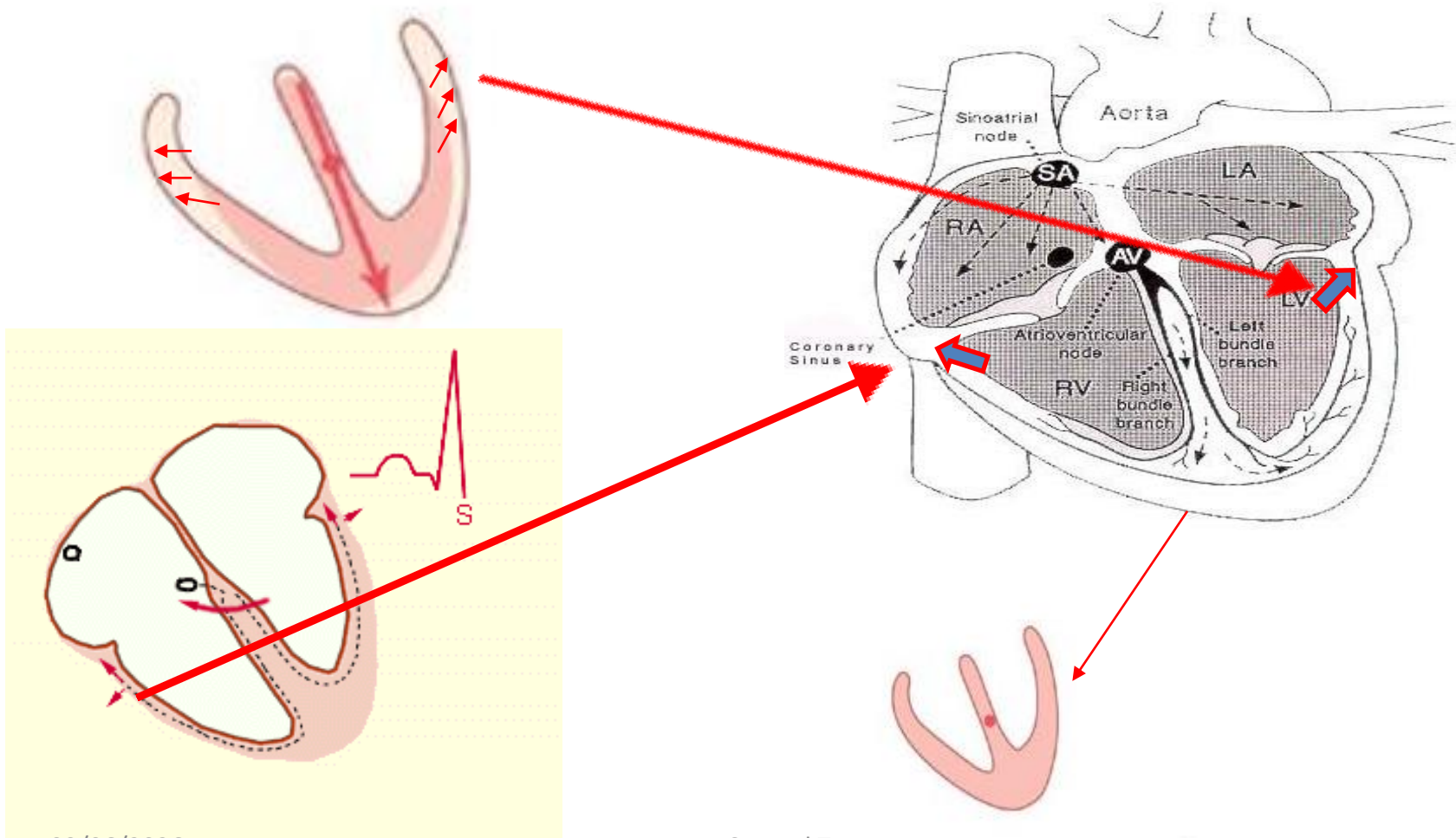
- Next, depolarization spreads along the *endocardial surfaces of the rest of the two ventricles*.
 - First the apex and later the base of the ventricle.
 - The apex of the ventricle produces a depolarizing vector moving in the direction of **down ward left and right**.



Vectorial Analysis of the Normal Electrocardiogram

- As the depolarization wave moves toward the apex more muscle is depolarized and less remains polarized especially toward the **epicardium at the base of the ventricles.**
 - The right ventricle having a lower tissue content so it **depolarizes more rapidly.**
 - But small area at the base of the left ventricle that remains polarized (Not depolarized).
 - As a result the depolarizing vector may move forward to the left ventricle to depolarized the myocardial then to the epicardial cells.
 - The depolarizing vector is moving ***upward direction to the electronegative node.***

Vectorial Analysis of the Normal ECG



Vectorial Analysis of the Normal ECG

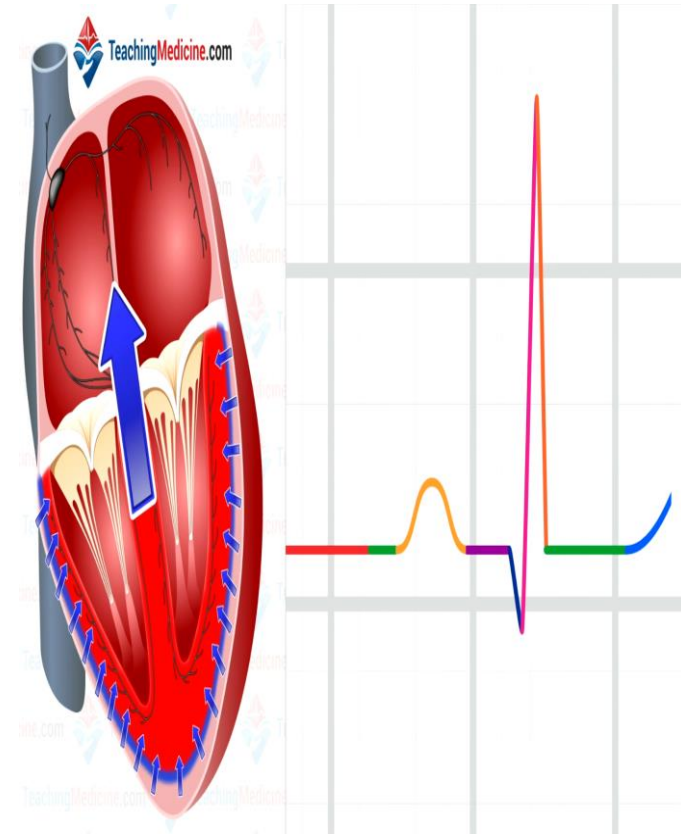
- As the remaining ventricular base muscle is being depolarized, a **downward deflection wave is produced**.
- Downward deflection is B/c of the depolarizing force (positive vector) move **up ward to the base** (away from the positive electrode) toward the negative electrode.
- As the ventricular mass completely depolarizes no other dipole is formed and the recorder returns to the **zero mark neutral point**.

Vectorial Analysis of the Normal ECG

- *Repolarization of the ventricles runs from epicardium to endocardium.*
- This is the reverse of what happens in depolarization.
- The principle in the myocardial cell is some what different.
 - *Cells which are depolarized last repolarized first.*
- *This occurs because the durations of the **plateau phases** of the action potentials in the individual cells is not the same.*
- The first ventricular myocyte to depolarize has the longest plateau.

Vectorial Analysis of the Normal ECG

- As the wave of repolarization moves from *epicardium to endocardium* more muscle is repolarized.
- When about half of the cardiac muscle is repolarized, the dipole or electrical potential vector is highest.
 - The vector is a **negative potential**.
- As more muscle repolarizes the dipole diminishes in magnitude and finally disappears leaving the heart repolarized and at the zero potential level.



Vectorial Analysis of the Normal ECG

- The magnitude of the recorded voltage in the ECG is directly proportional to the amount of tissue being depolarized.
- Therefore, the largest recorded wave form, the QRS complex, reflects the depolarization of the ventricles.
 - ❖ *Furthermore, since left ventricular mass is significantly greater than right ventricular mass, more of the QRS complex reflects the depolarization of **the left ventricle**, and **orientation of the mean electrical axis is to the left of the interventricular septum.***

Vectorial Analysis of Potentials Recorded in different Leads

- Current flows mainly downward from the base of the ventricles toward the apex than in the upward direction.
- This is shown by the **summated vector** (long black arrow) of the potential generated.

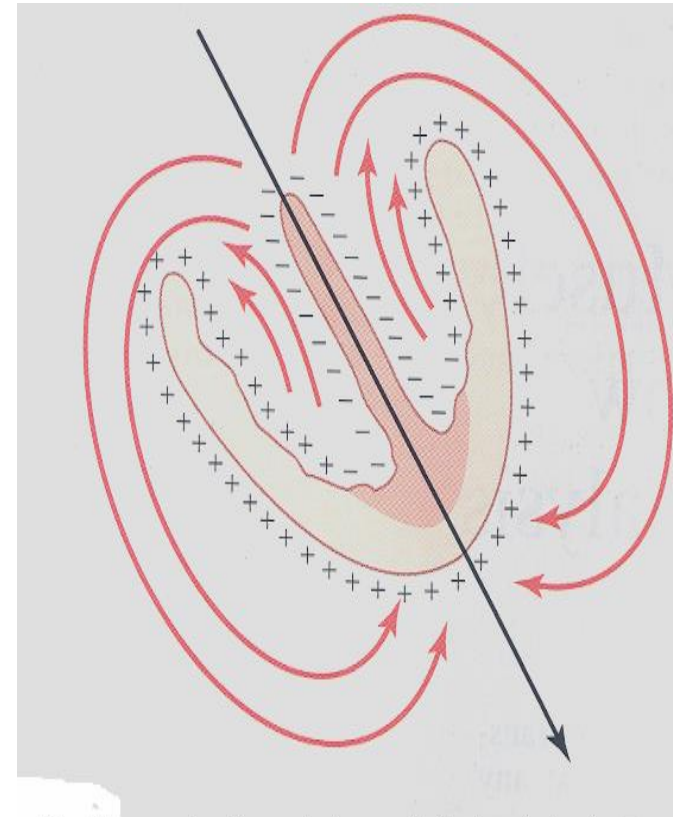


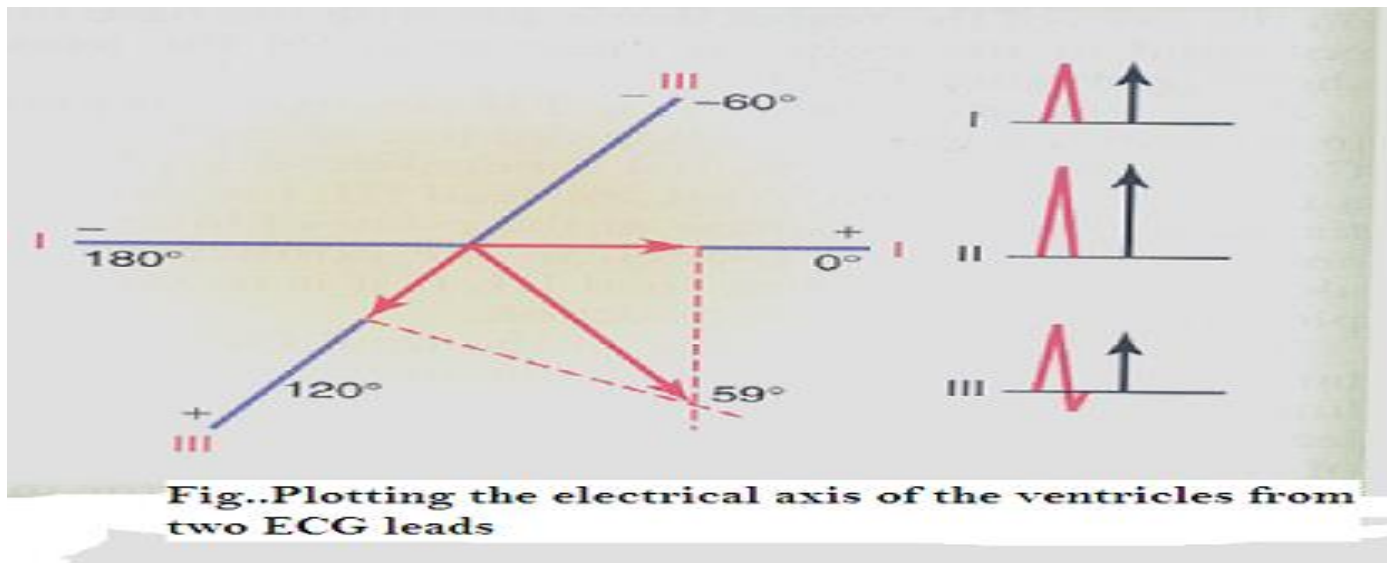
Fig..Mean vector through the partially depolarized ventricles

Vectorial Analysis of Potentials Recorded in different Leads

- When we use ***vector and axis together*** we will be able to determine the potentials recorded in ECG of each lead.
- *If we know the direction and amplitude of instantaneous mean vector of heart, by using axis of limb leads we can draw vectors and know about potential and direction of deflection in each limb lead*
- Negative potential subtracted from positive potential.

Vectorial Analysis

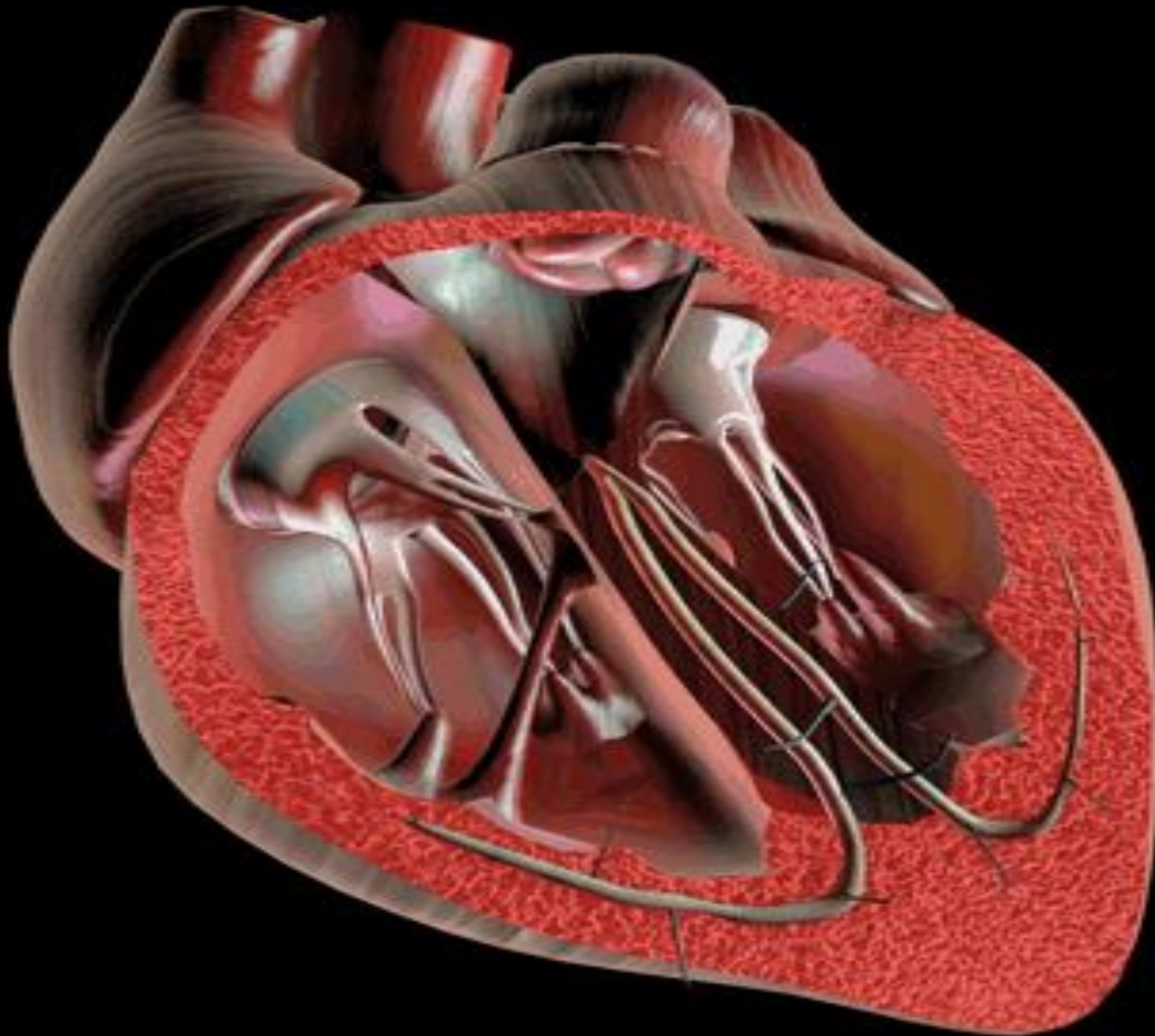
- Vectors of L I and L III(+ or -) plotted along lines depicting the leads and with the base at the point of intersection.
- Actual vector determined by drawing perpendicular lines for the apices of two vectors of L I and L III.
- *Point of intersection of two perpendicular lines, by vectorial analysis = Apex of the actual mean QRS vector in the ventricles.*



...cont'd

- *The mean QRS vector in the frontal plane is normally b/n -30° and $+100^{\circ}$.*
- Mean QRS vectors:
 - greater than $+100^{\circ}$ are said to represent **right axis deviation**,
 - those less than -30° are designated as **left axis deviation**.
- Mean vectors for ECG deflections other than the QRS, for e.g. mean T-wave vectors, can be calculated.

Cardiac Cycle



Cardiac Cycle

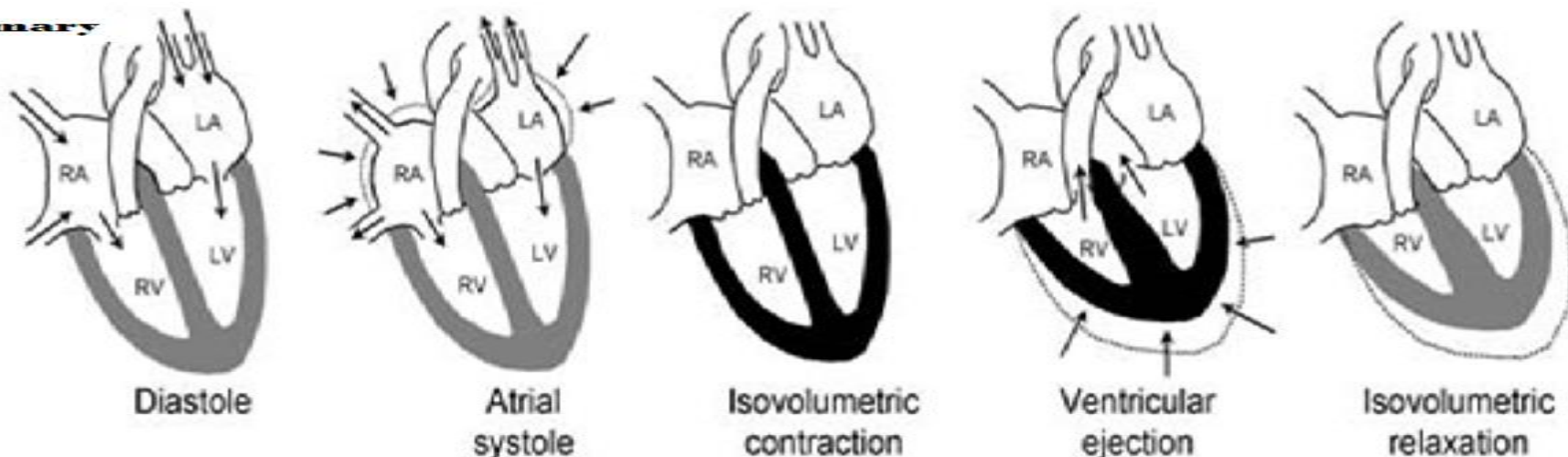
- Activities in the heart in a single beat
- Contraction (systole) and relaxation (diastole) of cardiac chambers
- A single cardiac cycle comprised of
 - Atrial systole and atrial diastole
 - Ventricular diastole + ventricular systole
- 75 cycles completed per minute
- Duration of each cycle = 0.8 second
 - Ventricular diastole = 0.5 second
 - Ventricular systole = 0.3 second

Means of exploring activities accomplished in a cardiac cycle

1. *Auscultation of heart sounds, phonograph*
2. *ECG tracing*
3. *Measuring aortic pressure*
4. *Measuring atrial pressure*
5. *Measuring ventricular pressure and volumes*

Phases of the Cardiac Cycle

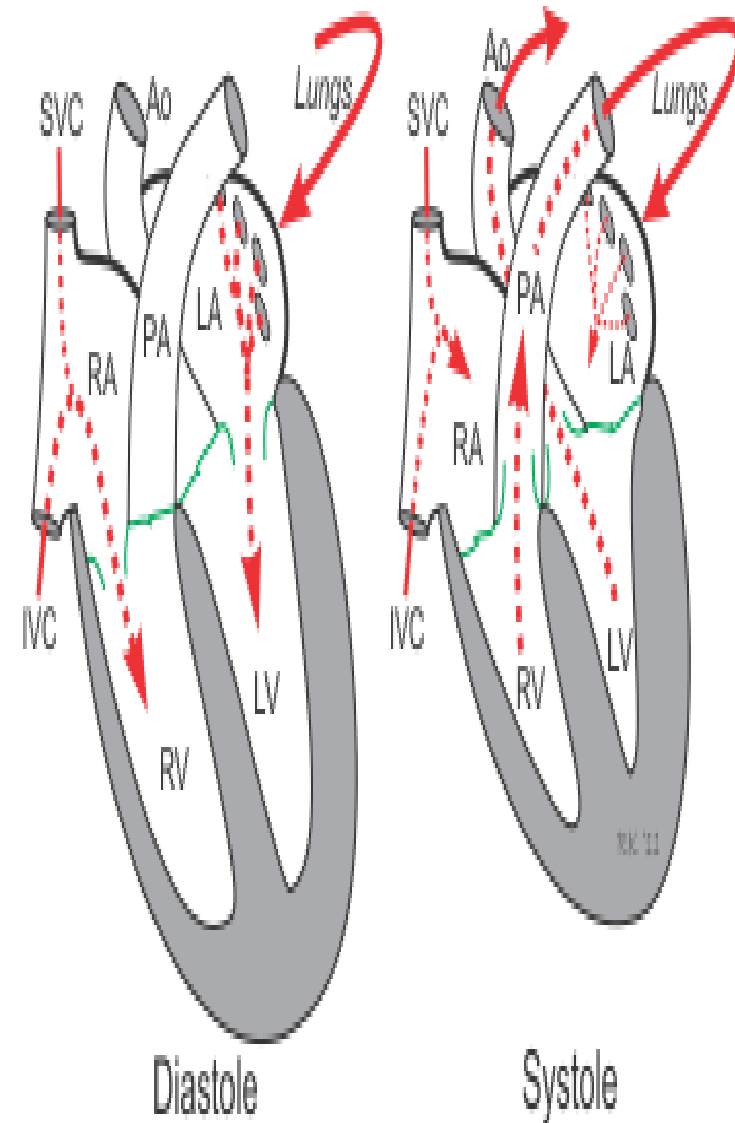
Summary

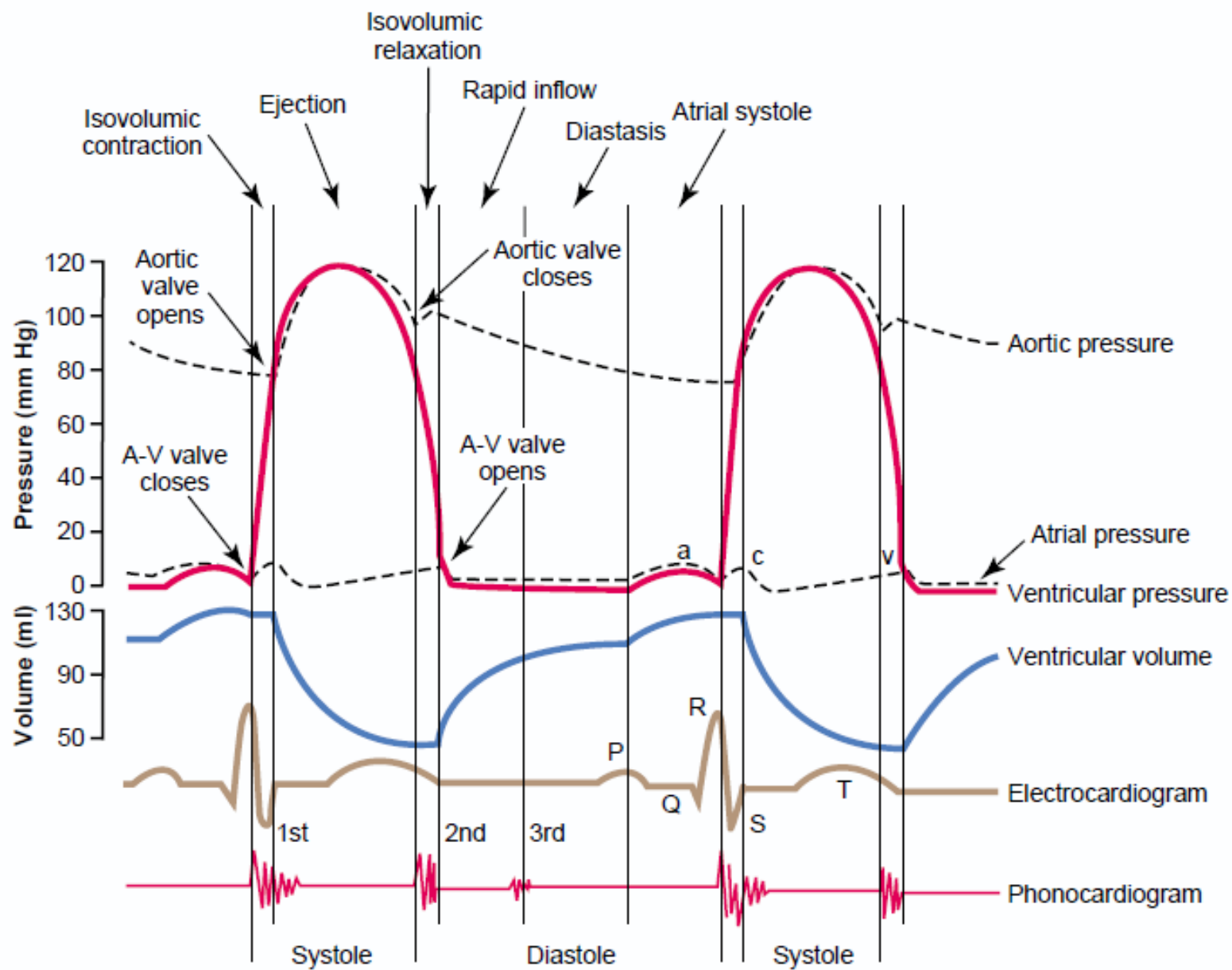


Ventricular Systole	Duration (approximate)
➤ Isovolumetric contraction	0.05 sec
➤ Rapid ejection	0.10 sec
➤ Slow ejection	0.15 sec
Ventricular Diastole	
➤ Isovolumetric relaxation	0.10 sec
➤ Rapid filling	0.10 sec
➤ Slow filling (Diastasis)	0.20 sec
➤ Filling during Atrial Systole	0.10 sec

Phases of the Cardiac Cycle

- Includes the following events:
 - **Electrical events** of summated ECG voltage changes-P wave, QRS complex, T wave and U wave in some cases
 - **Mechanical events** of myocardial diastole and systole with opening and closing of cardiac valves, pressure and volume changes, heart sounds, atrial (jugular) a,c,v, waves.





Phases of ...cont'd

A. Phase of Diastole (Atrial and ventricular diastole):

- Atrial pressure rises before the tricuspid valve opens during diastole. This is mirrored by **v-wave**
- Atria and ventricles are relaxed.
- As pressure falls below that in the atrium, it leads to opening of AV valves and closing of semi-lunar valves
- Blood flows into the heart filling both atrial and ventricular chambers throughout diastole.
- When pulse rate is slow and diastole is long, inflow is slowed in the latter part of diastole → a period of **diastasis**.

Phases of ...cont'd

- The 3rd heart sound begins with end of rapid inflow and ends with beginning of diastasis.
- Proto-diastole + diastasis = 70% of inflow
- About 70% of the ventricular filling occurs passively.
- The remaining 30% of filling occurs by active atrial systole
- P-wave formation begins with diastasis and ends with atrial systole.
- The beginning of the 1st heart sound and the end of QRS complex coincide with isovolumetric or isometric contraction (0.02-0.06 sec).
- 2nd heart sound occurs with closure of semilunar valves and during isometric relaxation (0.09-0.12 sec)

Phases of ...cont'd

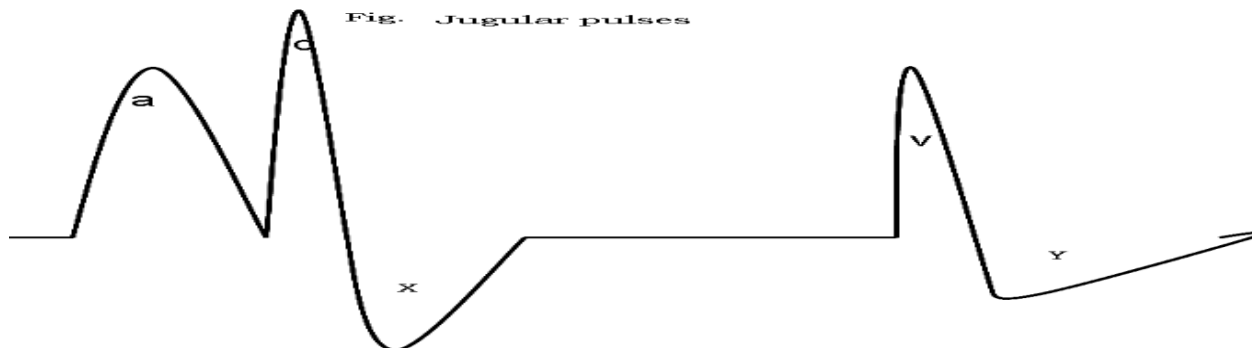
B. Phase of Systole

a. Atrial systole (contraction):

- Propels an additional blood of about 30% of EDV into ventricles
- Preceded by P-wave
- EDV = 120 mL
- Forms a-wave, formed by atrial systole.
- This is one of the three atrial pressure waves (a,c,v) transmitted to thoracic and neck veins (Fig.below)
- ↑ ventricular volume
- ↑ ventricular pressure (atrial kick)

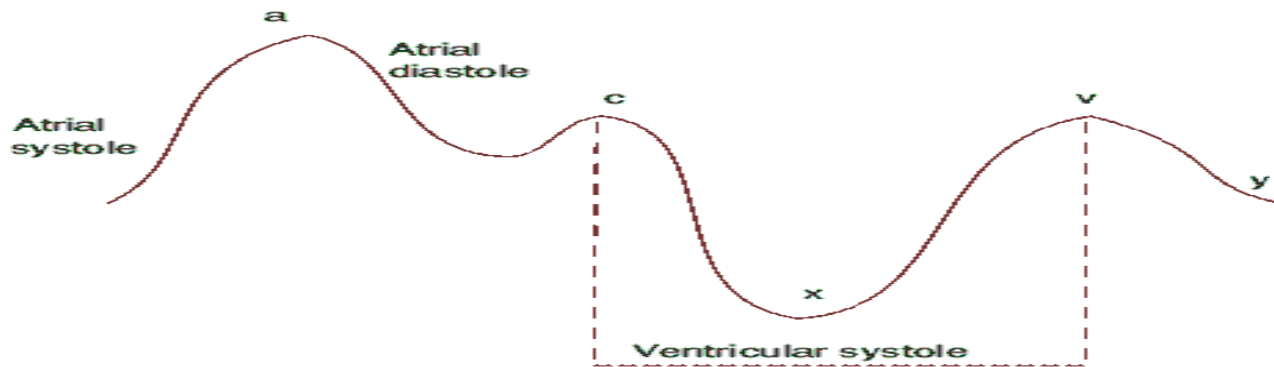
Jugular Venous Pulse Waves

- **a-wave** is due to atrial systole which causes increase in the intra-atrial pressure
- **C-wave** occurs during the isometric contraction of the ventricles due to bulging of the tricuspid valve
- **V-wave** is due to accumulation of blood during late part of the ventricular systole or protodiastolic phase and isometric relaxation of the ventricles



Jugular Venous Pulse Waves

- **X-wave** is due to downward displacement of the A-V valve (tricuspid valve) and the AV ring during maximum ejection phase of ventricular contraction
- **Y-wave** is due to opening of the AV valve (tricuspid valve) during maximal ventricular filling phase



Jugular Venous Pulse Waves

- “a-c” interval measures the time of conduction of the cardiac impulse from the right atrium to the ventricles
- The a-c interval corresponds to the P-R interval of the ECG
- The “a-c” interval is prolonged in cases of delayed conductivity in the AV bundle which is an early sign of heart block
- In partial heart block, the number of “a” waves is greater than the number of the “c” or “v” waves.
- In atrial fibrillation, the “a” wave is absent.

Phases of ...cont'd

b. Ventricular Systole

- During this period:
- Isovolumetric contraction (**0.02-0.06 sec**) occurs
- i.e, there will be:
 - No change in ventricular volume or length of ventricular muscle fibers
 - Rise of BP in left ventricle from 4 to 80 mmHg
 - Closed AV valves
 - Closed S-L valves
 - Aortic and pulmonary pressures are at their lowest levels

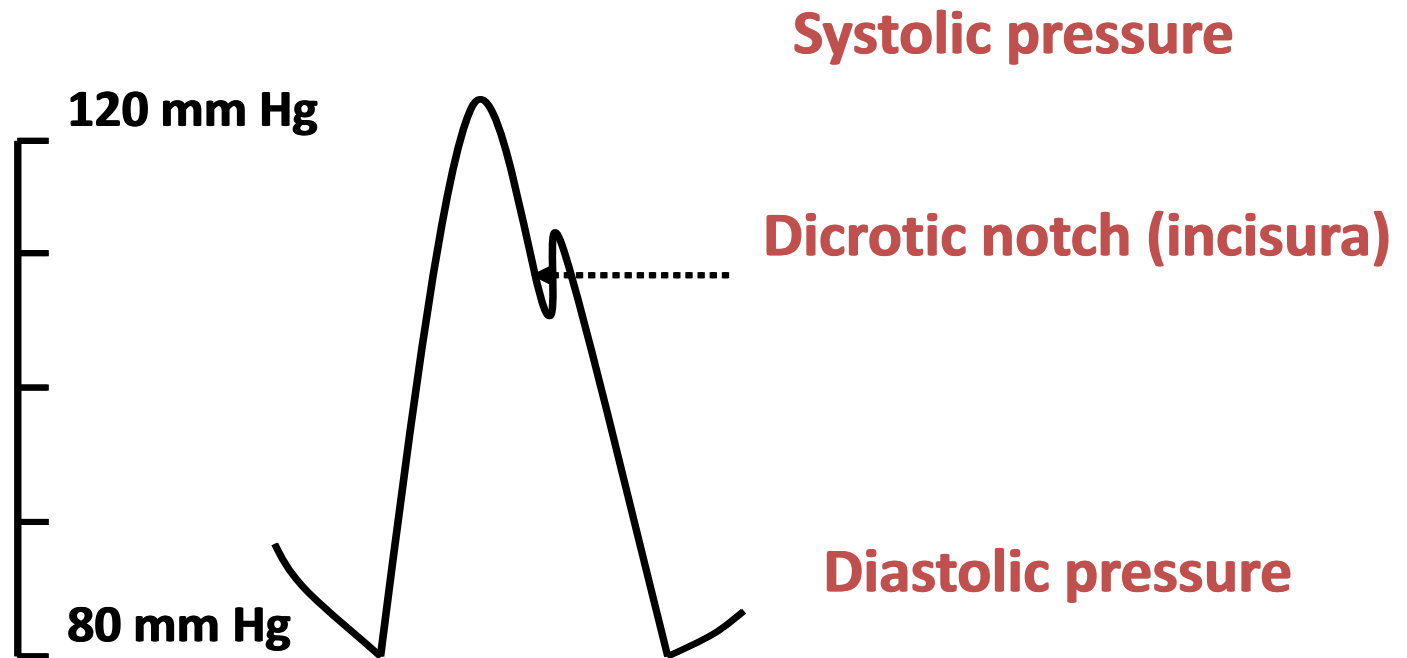
Phases of ...cont'd

- **c-wave** as transmitted manifestation of the rise in atrial pressure is produced by the bulging of tricuspid valve into atria during isovolumetric contraction (**ventricular systole**).
- **c-wave** is synchronous with pulse wave in carotid artery, and hence the use of the letter “c”,
- During isovolumetric contraction part of the first heart sound develops, QRS complex develops fully and Ventricular pressure rises from 4mmHg to 80 mmHg
- At the end of isovolumetric contraction:
 - S-L valves open
 - Ventricular pressure rises from 80 mmHg to 120 mmHg
 - **Rapid ejection period (0.10 sec) occurs...70% of SV**
 - **Slow (reduced) ejection (0.20 sec) follows,,30% of SV**
 - Isovolumetric relaxation follows

...cont'd

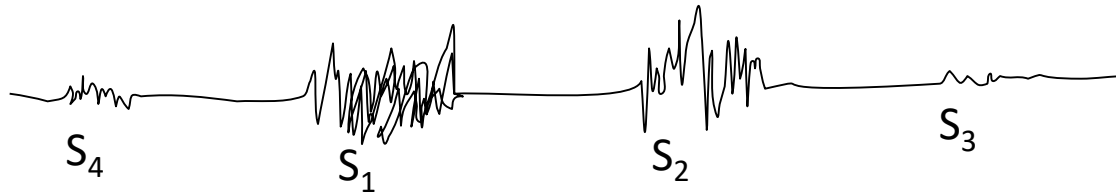
- $SV = 80 \text{ ml}$, i.e. ejection fraction $(80/120)$, $EF = 67\%$, $ESV = 120 - 80 \text{ ml} = 40 \text{ ml}$
- Ventricular volume goes down to ESV
- Most of the T-wave occurs during the ejection period
- Pressure in ventricle falls slightly below atrial pressure
- The aorta which being distensible **undergo recoil** and produce a **secondary pressure wave** in aortic curve with a notch between systolic and diastolic pressure curves.
- This is referred to as **dicrotic notch - a small oscillation** on the falling phase of the pulse wave (Fig. below) formed when the **aortic valve snaps shut** following full emptying of blood.

Aortic pressure



Heart Sounds

- Heart sounds (lubb-dubb) are associated with **closing** of heart valves
- 4-separate audible heart sounds (S_1 , S_2 , S_3 and S_4)
- Means of identification
 1. Auscultation: direct/immediate auscultation
 Stethoscope mediated auscultation
 2. Phonocardiographic based recording



- S_1 : is always audible, has a LUBB-sound
- S_2 : is always audible, has DUBB-sound
- Continuous heart sound: Lubb-Dubb, Lubb-Dubb
- S_3 : is audible in children and in adults during exercise
- S_4 : is audible very rarely

Phases of the Cardiac Cycle

